

Project Scheduling

Example

PROJECT SCHEDULING

- ❑ It is part of **project management** within the *Planning* phase of the Systems Development Life Cycle.
- ❑ **Project Scheduling:** Allocate resources to execute all activities in the project.



- ❑ **Project:** Set of activities or tasks with a clear *beginning* and *ending* points. The amount of available resources (time, personnel and budget) to carry out the activities is usually limited.
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CPM – CRITICAL PATH METHOD

Objectives:

- Establish **beginning, ending** and **duration** of each activity in the project.
 - Calculate overall completion time of the project given the amount of usually limited resources.
 - Determine the **critical path** and its **duration**.
 - Determine the **slack time** for all **non-critical activities** and the whole project.
 - Guide the allocation of resources other than time such as staff and budget.
-

CPM – CRITICAL PATH METHOD

Phase:

- 1. Define **activities** or **tasks** according to the project **objectives**.
 - A **task** is an individual unit of work with a clear beginning and a clear end.
 - Identify **precedence** relationships or **dependencies**
 - Estimate **time required** to complete each task.
 - Draw an **activity-on-arrow PERT** diagram inserting **dummy** activities if required.
 - Apply **CPM** to calculate *earliest* and *latest starting times*, *earliest* and *latest completion times*, *slack times*, *critical path* etc.
 - Construct a **GANTT** chart.
 - Reallocate resources and resolve if necessary.
 - Continuously monitor/revise the time estimates along the project duration.
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CPM – CRITICAL PATH METHOD

- ❑ It is determined by adding the times for the activities in each sequence.
- ❑ CPM determines the **total calendar time** required for the project.
- ❑ If activities outside the critical path speed up or slow down (within limits), the total project time does not change.
- ❑ The amount of time that a **non-critical** activity can be delayed without delaying the project is called **slack-time**.

CPM – CRITICAL PATH METHOD

ES	Activity	EF
LS	Duration	LF

- ES – Activity earliest start time
 - LS – Activity latest start time
 - EF – Activity earliest finishing time
 - LF – Activity latest finishing time
 - Slack Time – Maximum activity delay time
-

CPM – CRITICAL PATH METHOD

Step 1. Calculate ET for each node.

For each node i for which predecessors j are labelled with ES(j), ES(i) is given

by:

$$ES(i) = \max_j [ES(j) + t(j,i)]$$

where $t(j,i)$ is the duration of task between nodes

(j,i). Step 2. Calculate LT for each node.

For each node i for which successors j are labelled with LF(j), LF(i) is given

by:

where $t(j,i)$ is the duration of task between nodes (i,j).

$$LF(i) = \min_j [LF(j) - t(i,j)]$$

CPM – CRITICAL PATH METHOD

Step 3. Calculate slack time for each node.

An activity with **zero slack time** is a **critical activity** and cannot be delayed without causing a delay in the whole project.

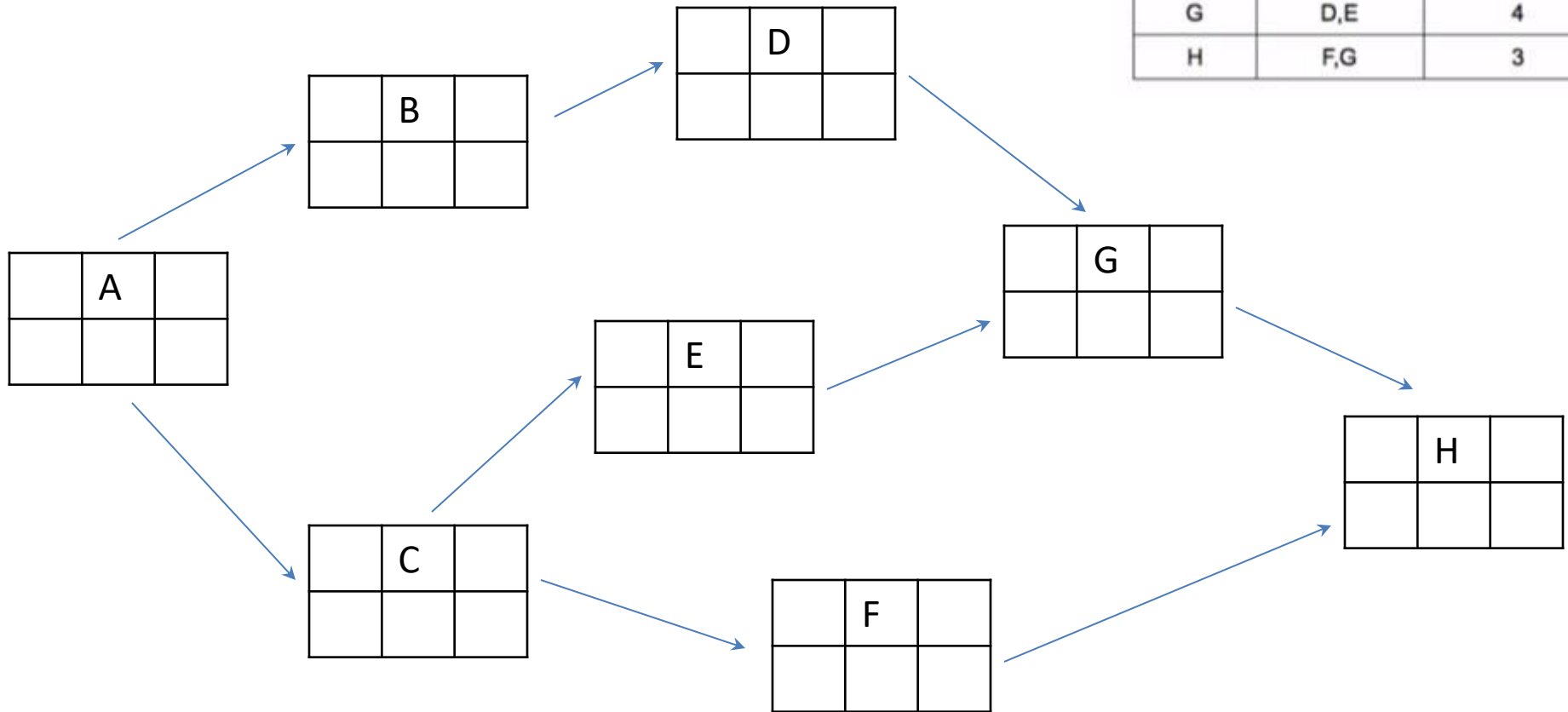
Total Float/ Slack Time: is the amount of time that an activity can be delayed from its early start date without delaying the project finish date.

Free Float: is the amount of time that an activity can be delayed without delaying the early start date of any successor activity.

ES	Activity	EF
LS	Duration	LF

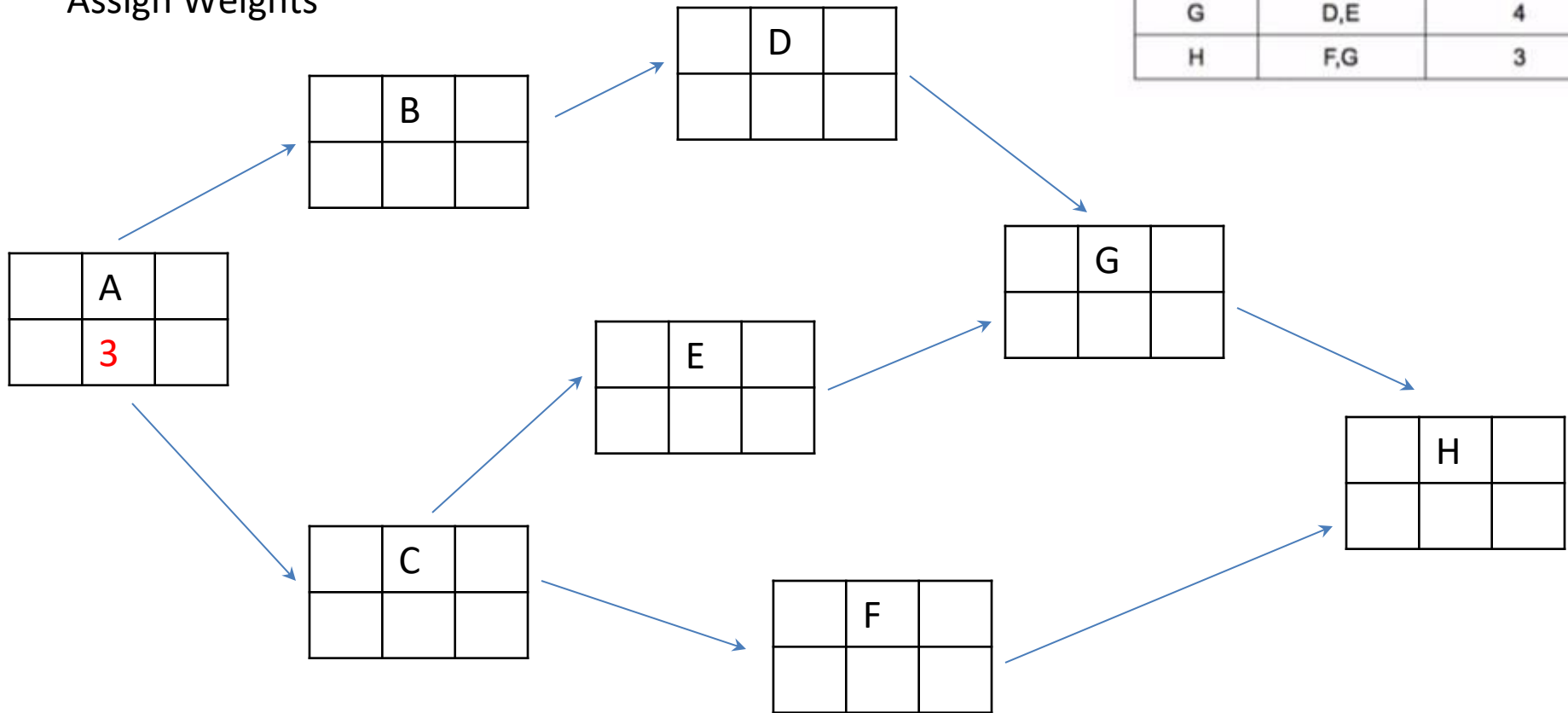
Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D,E	4
H	F,G	3

Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D,E	4
H	F,G	3



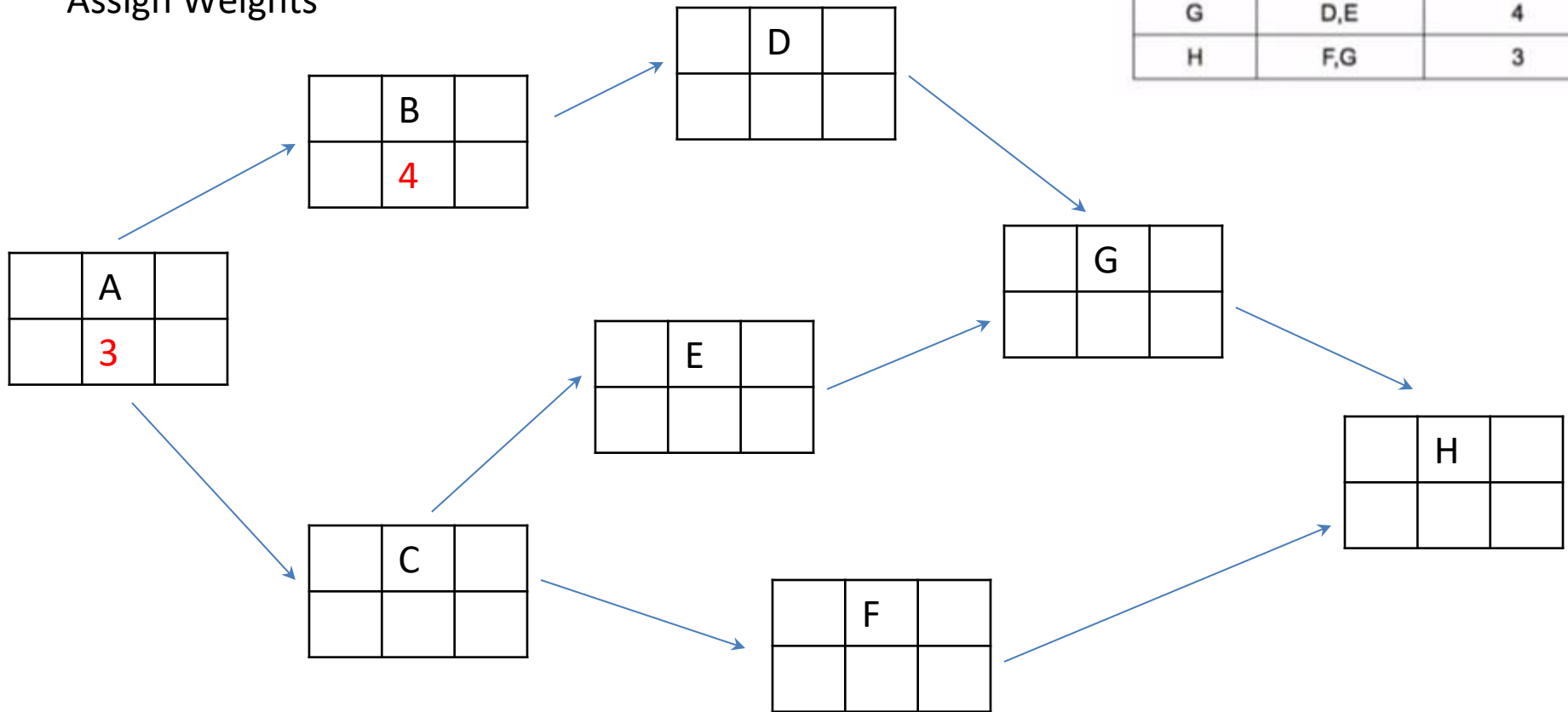
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Assign Weights



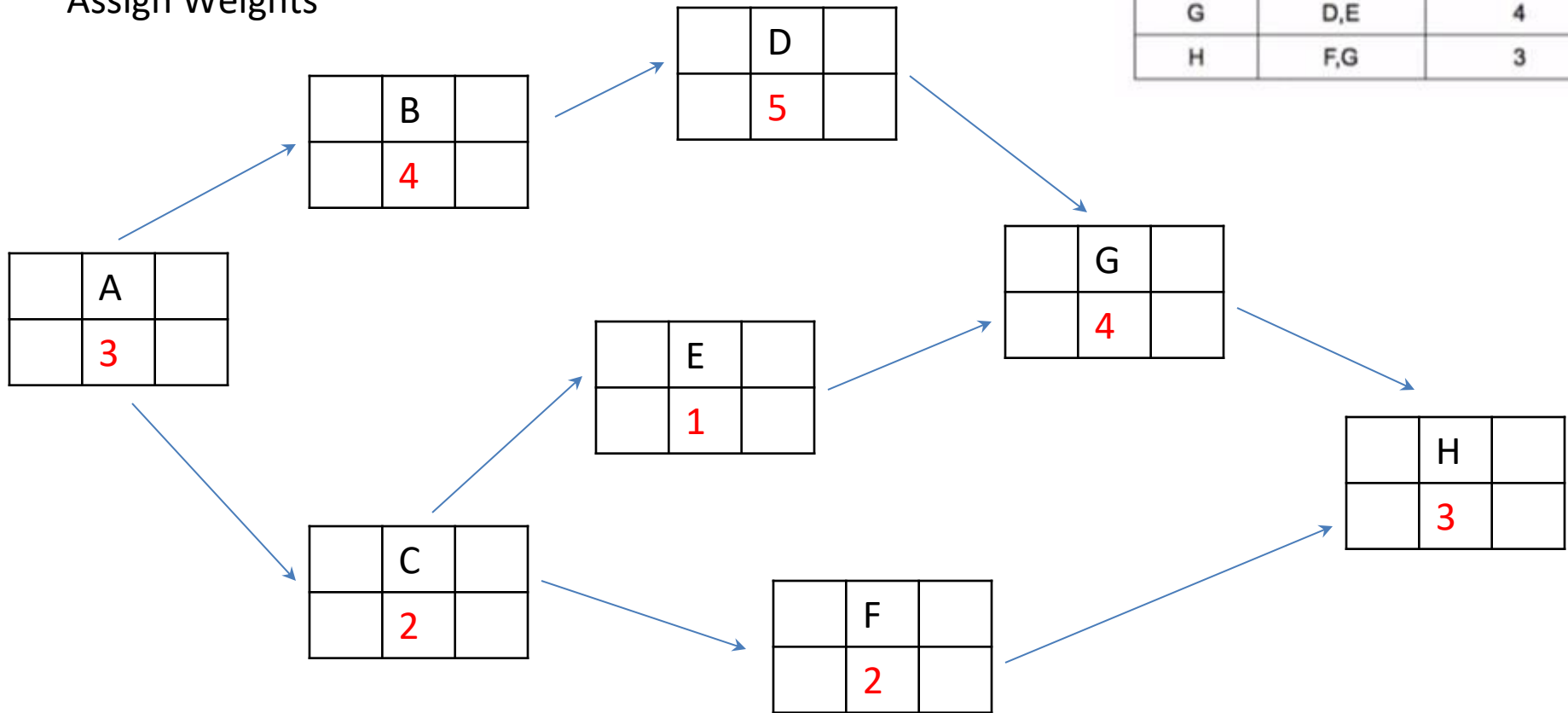
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Assign Weights



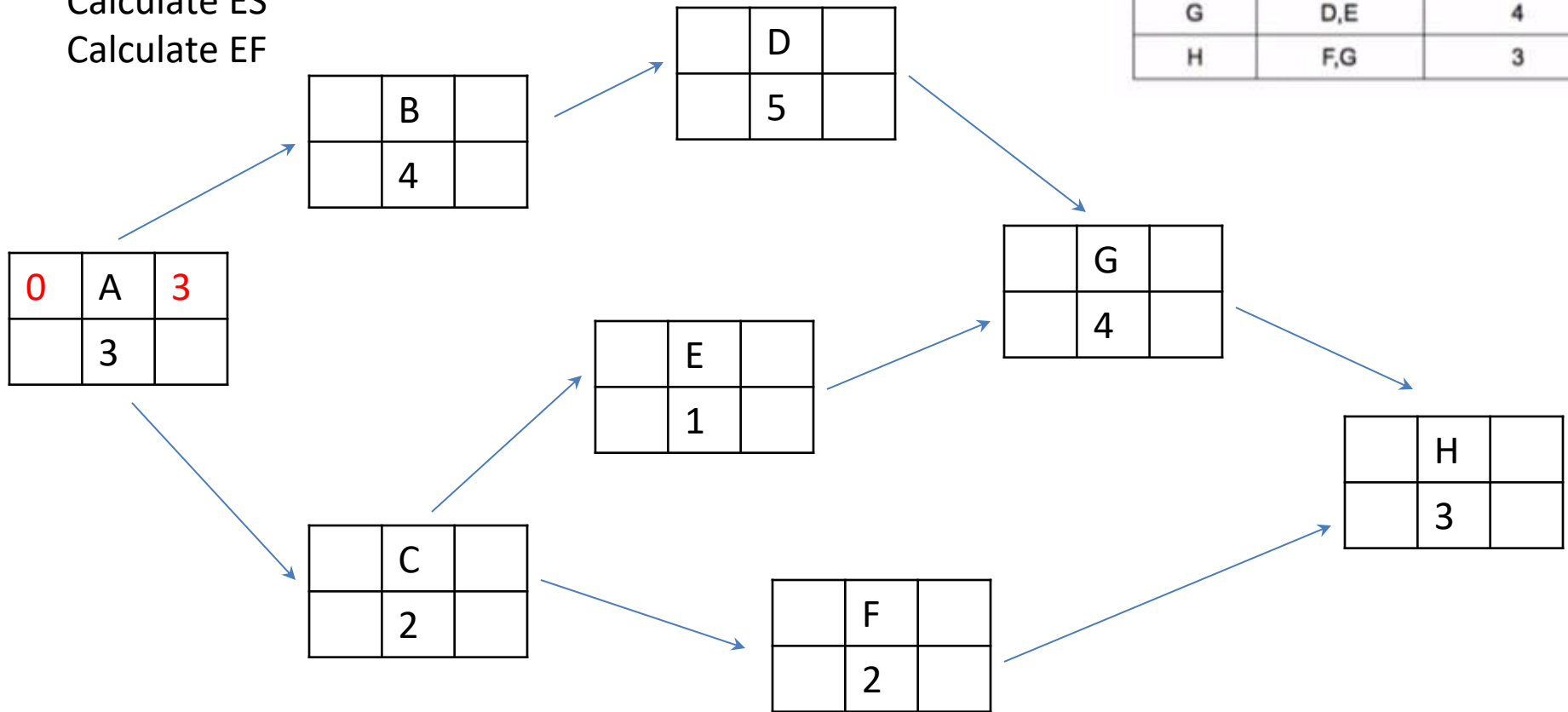
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Assign Weights



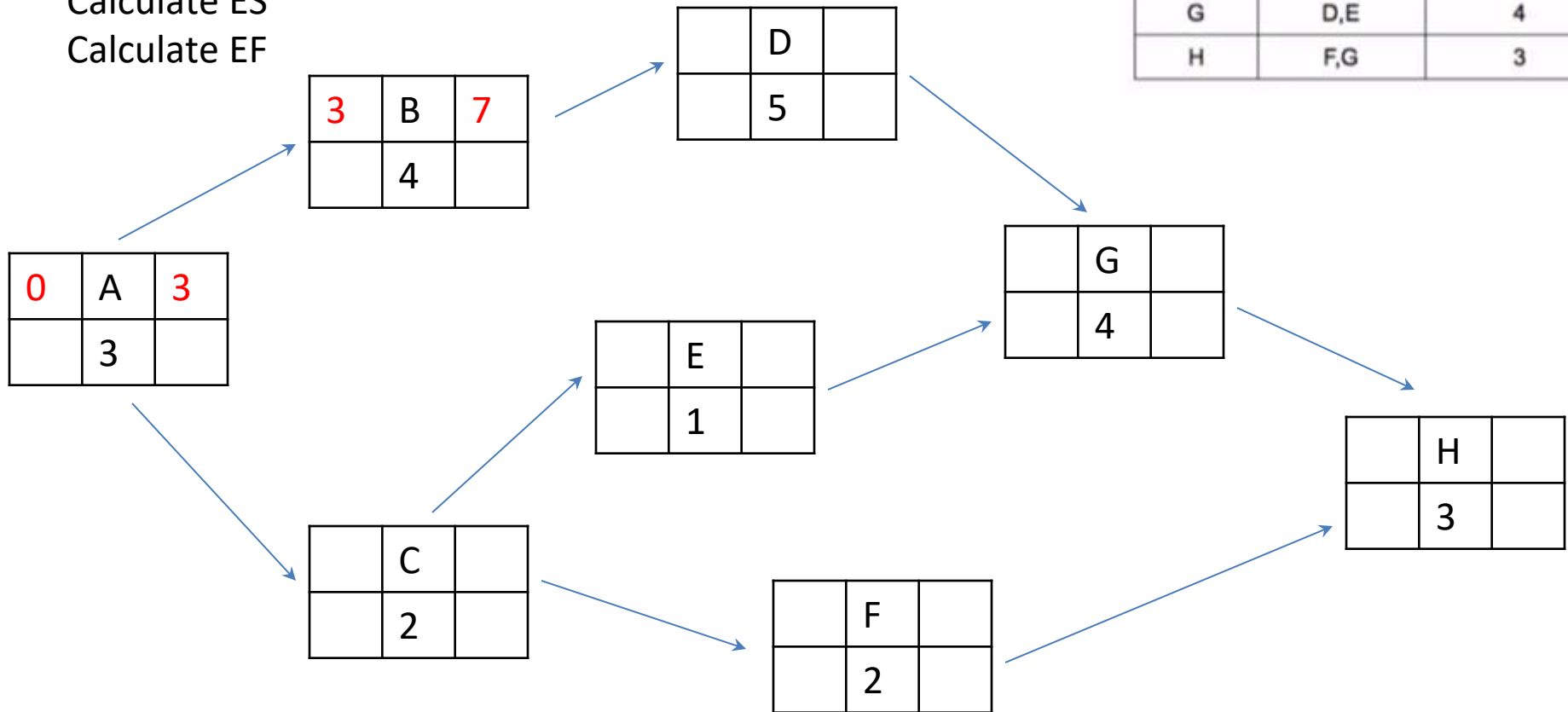
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D	B	5
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Calculate ES
Calculate EF



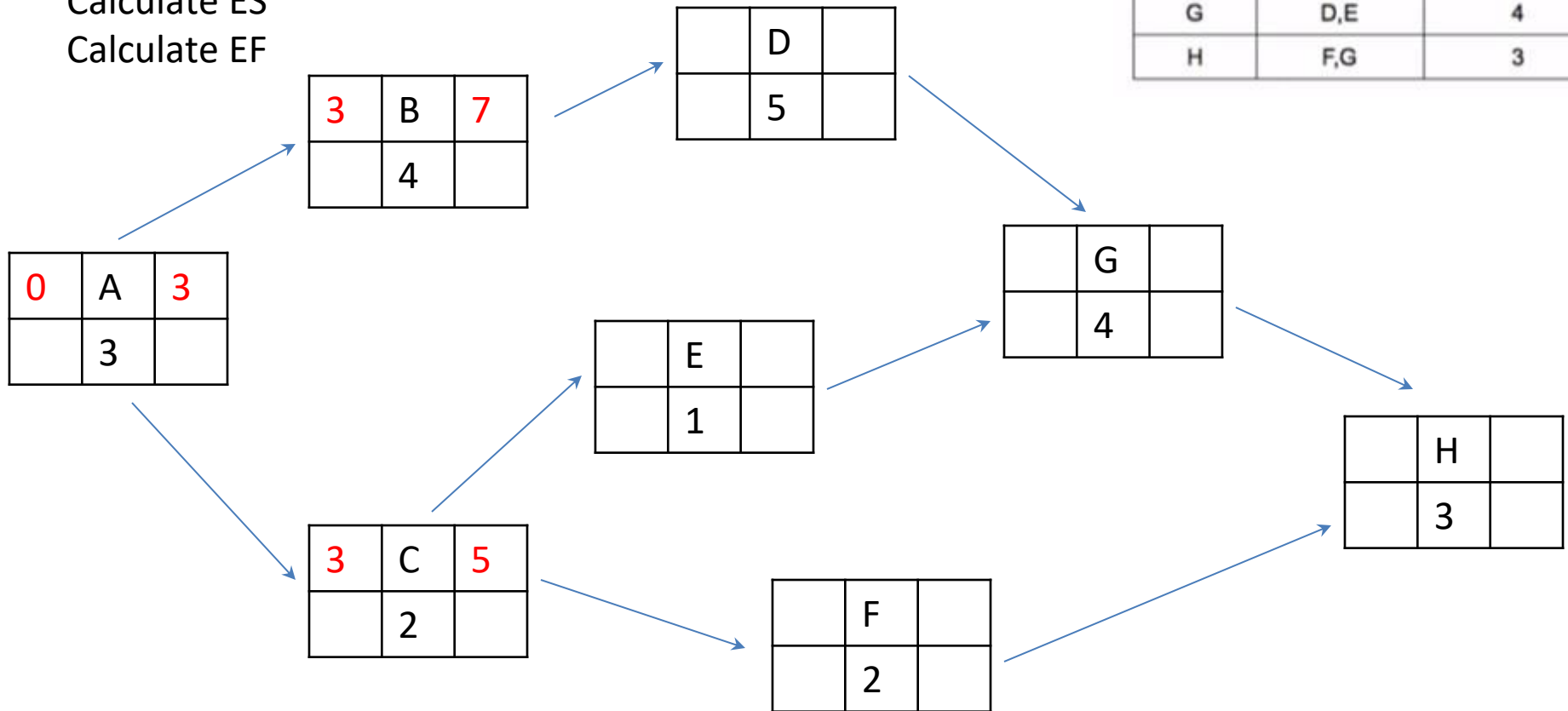
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D	B	5
E	C	1
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Calculate ES
Calculate EF



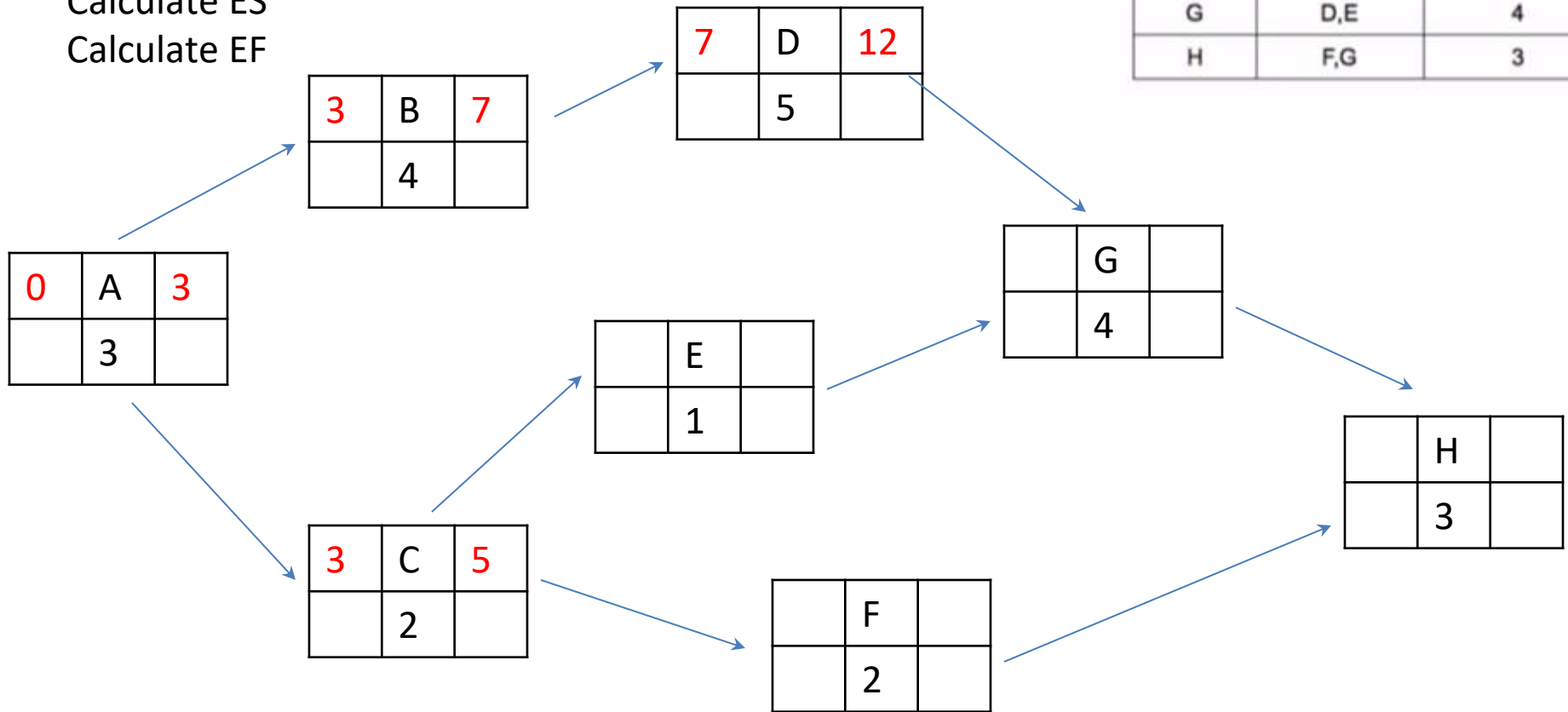
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Calculate ES
Calculate EF



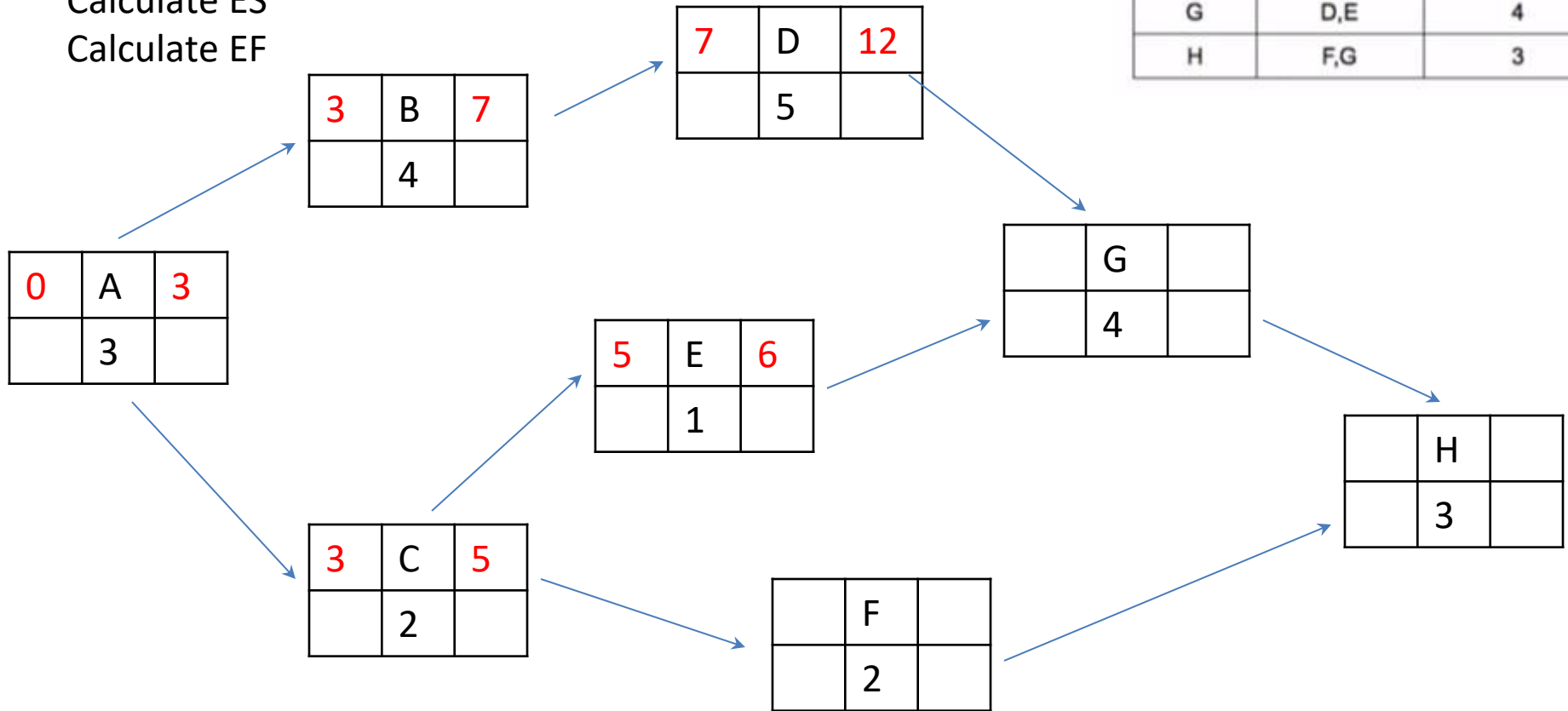
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Calculate ES
Calculate EF



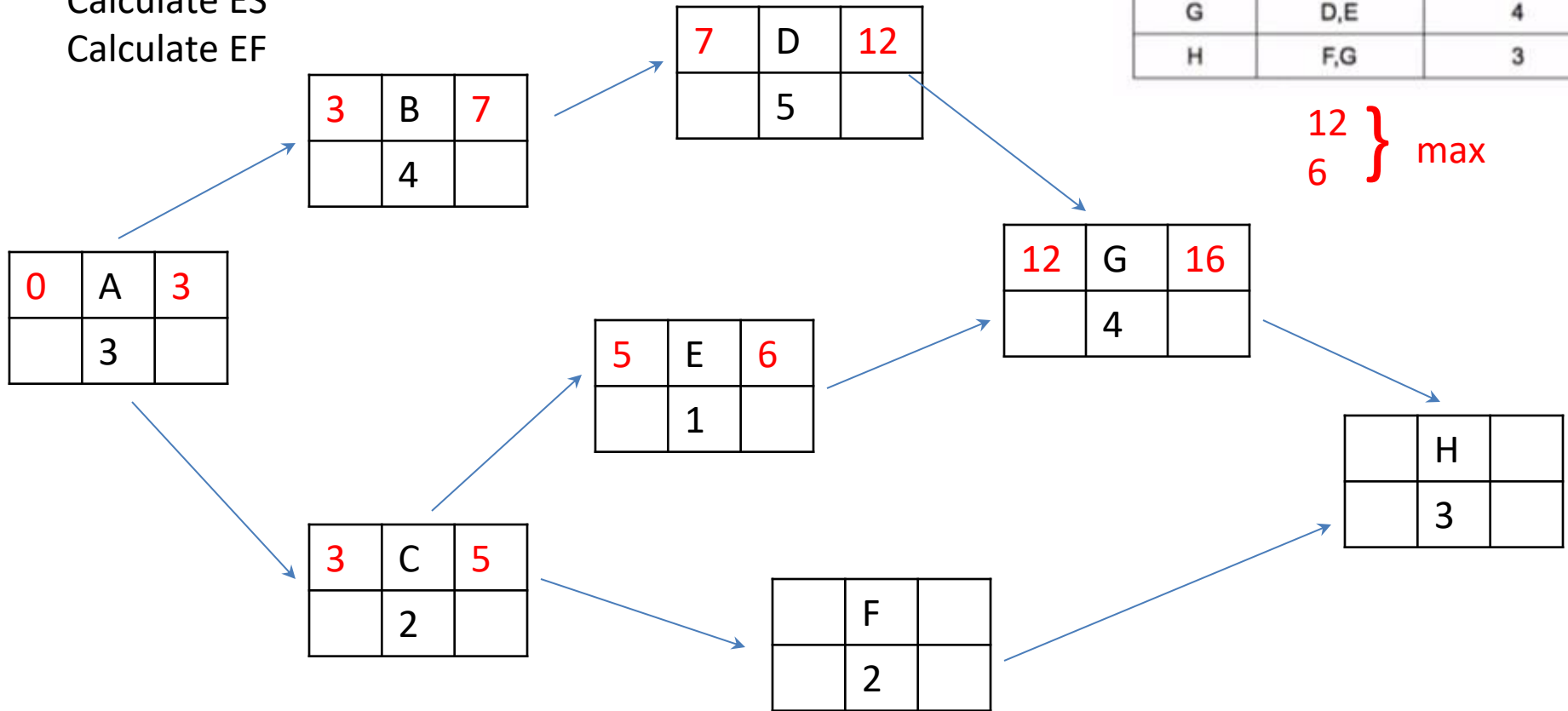
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Calculate ES
Calculate EF



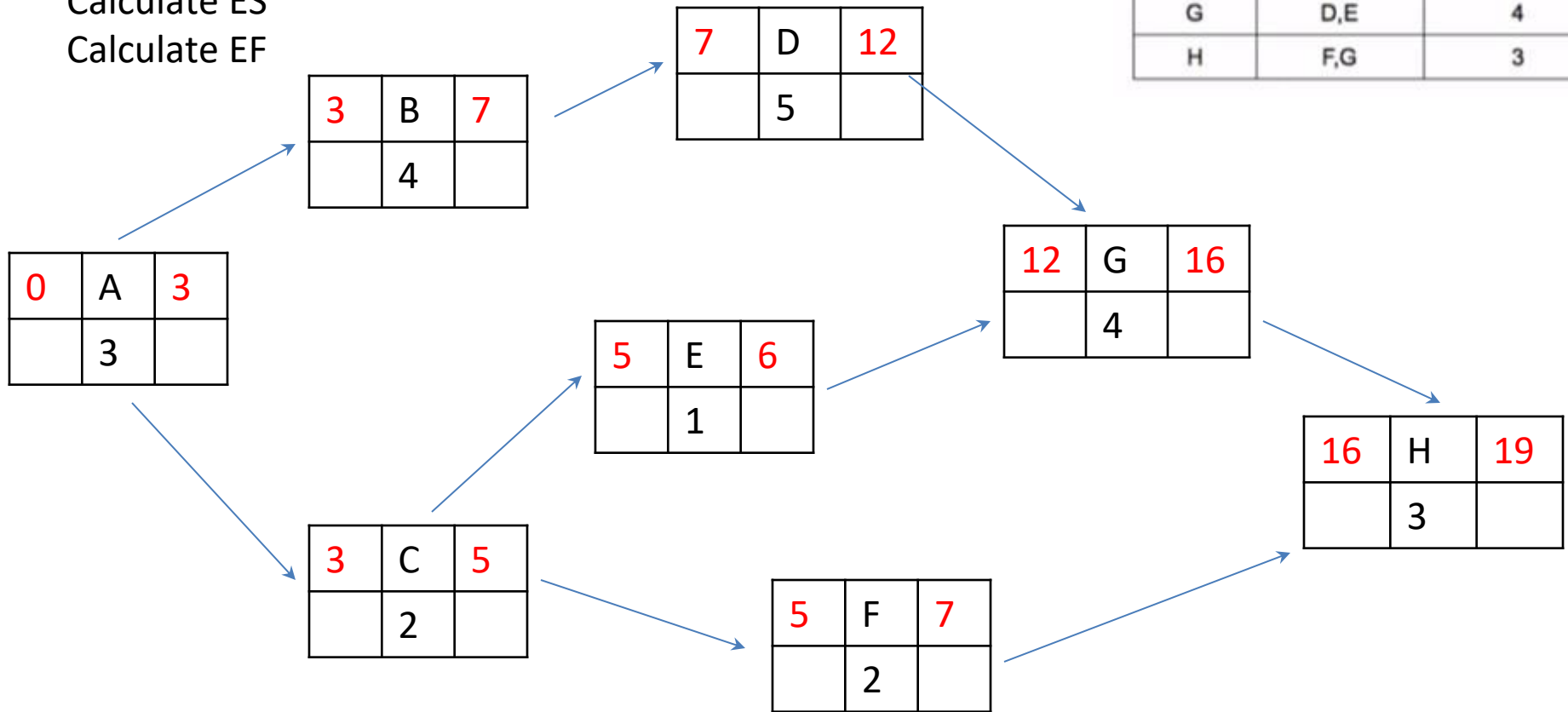
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Calculate ES
Calculate EF



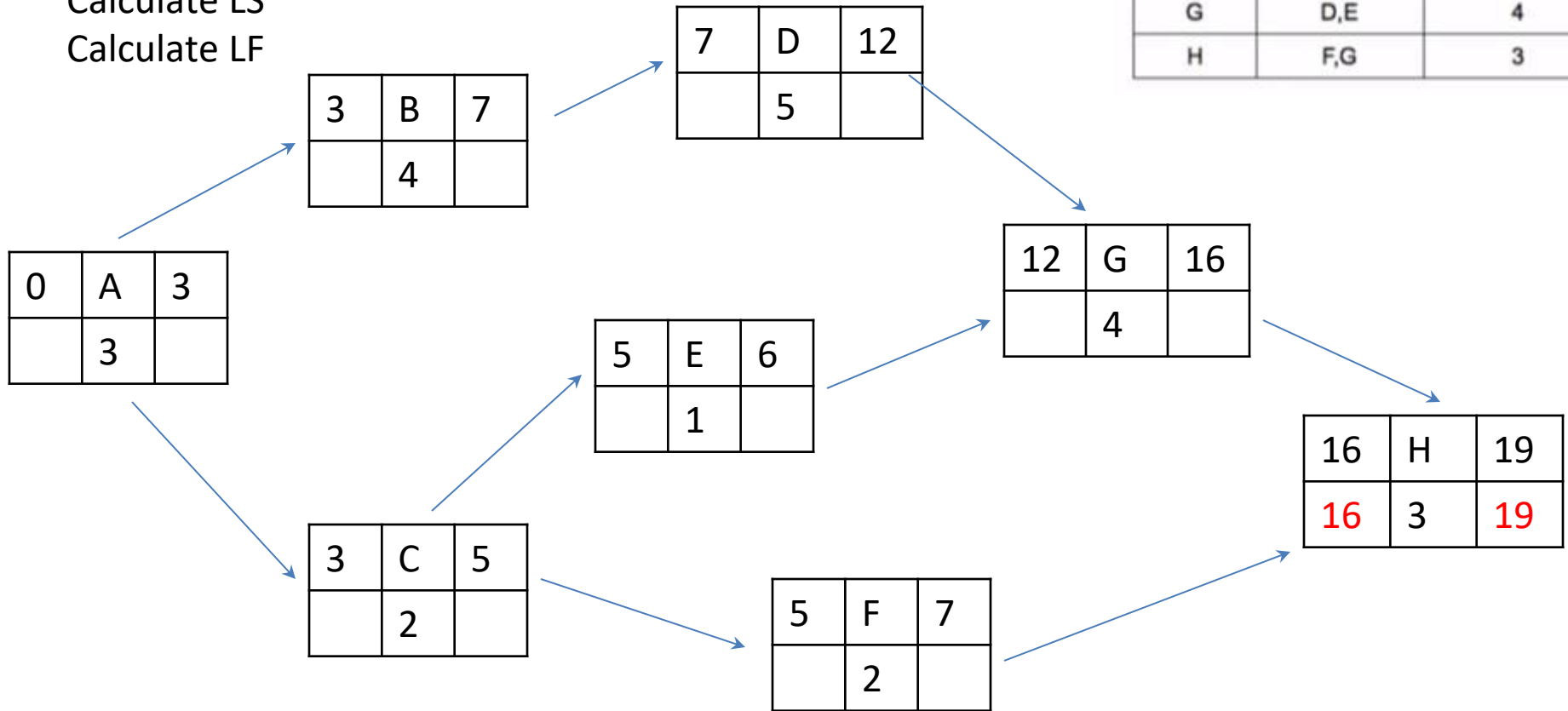
Activity	Predecessor	Duration (days)
A	-	3
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Calculate ES
Calculate EF



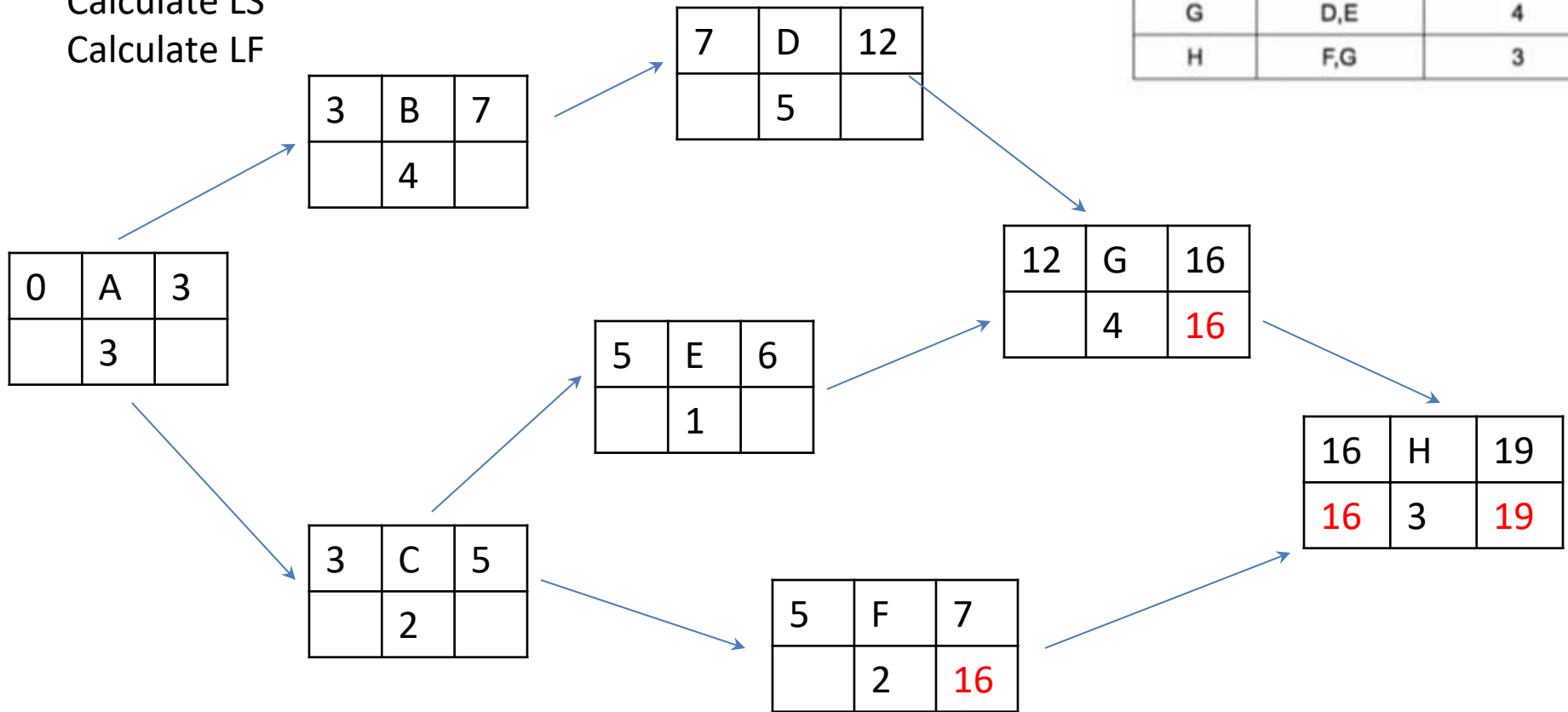
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Calculate LS
Calculate LF



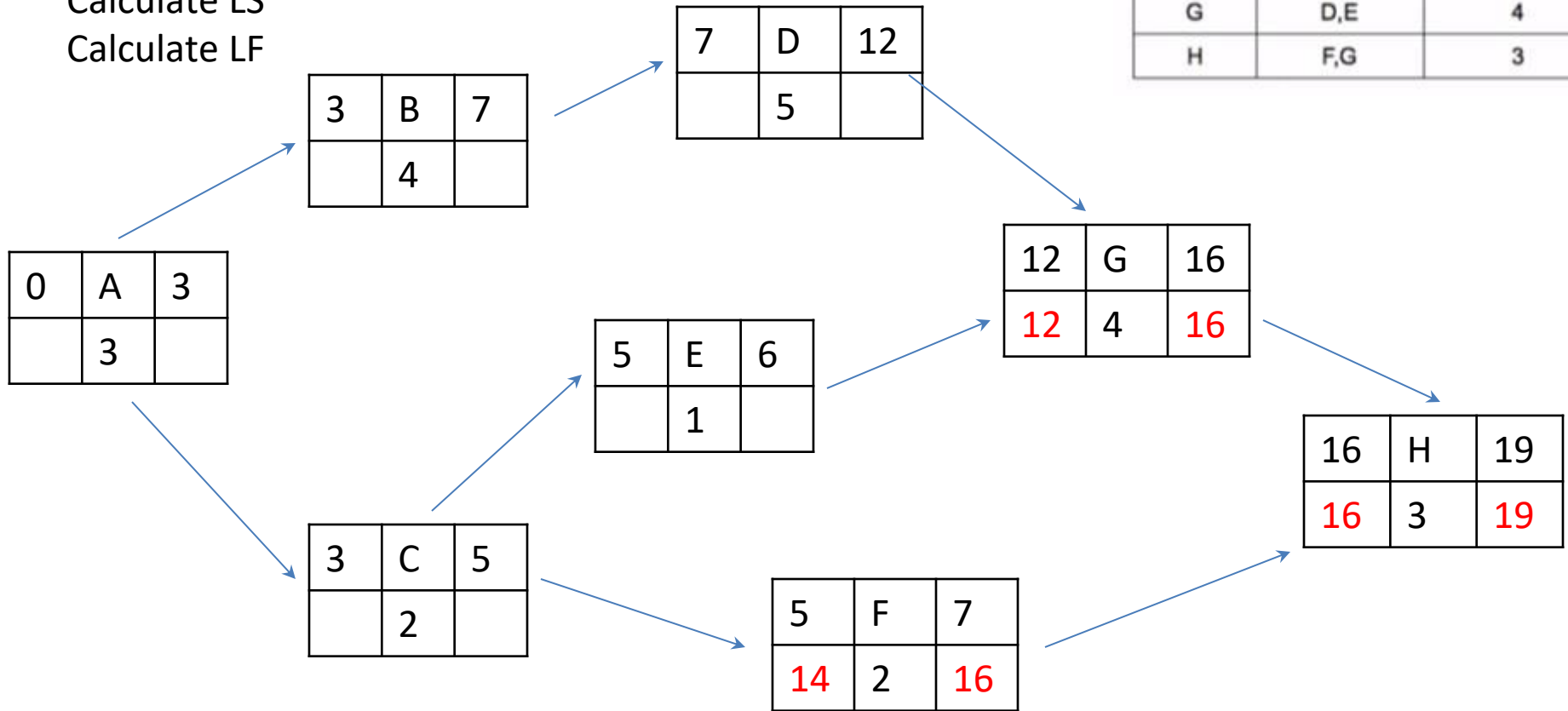
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Calculate LS
Calculate LF



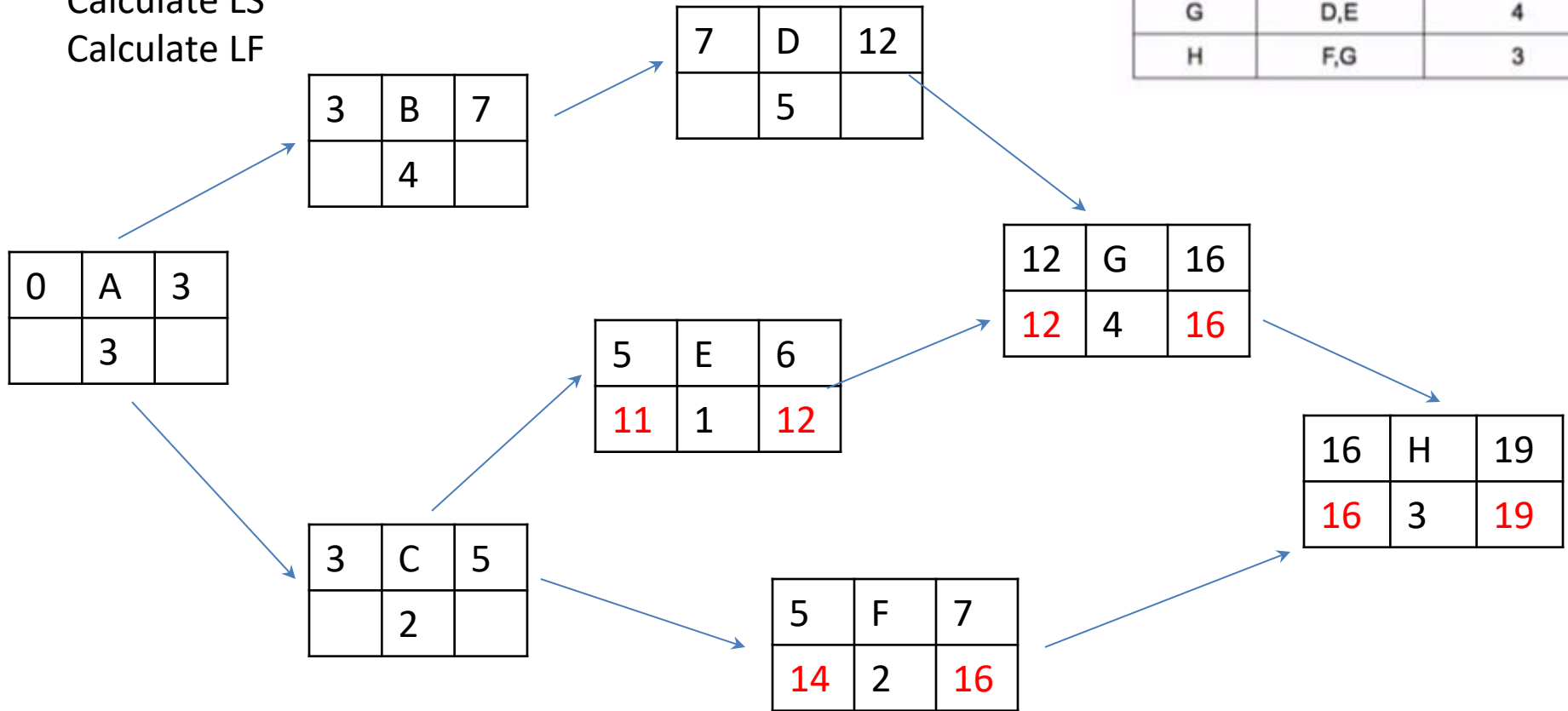
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Calculate LS
Calculate LF



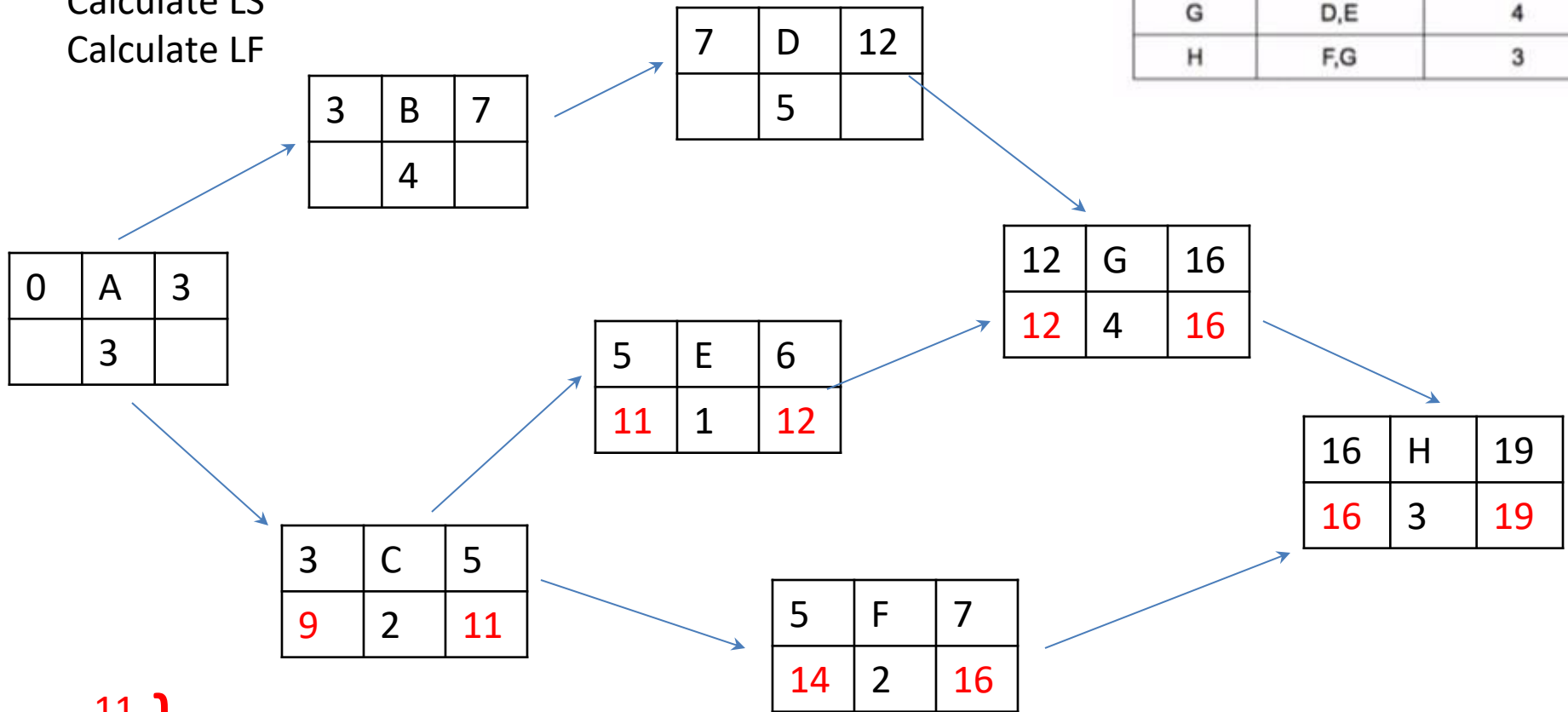
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A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
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Calculate LS
Calculate LF



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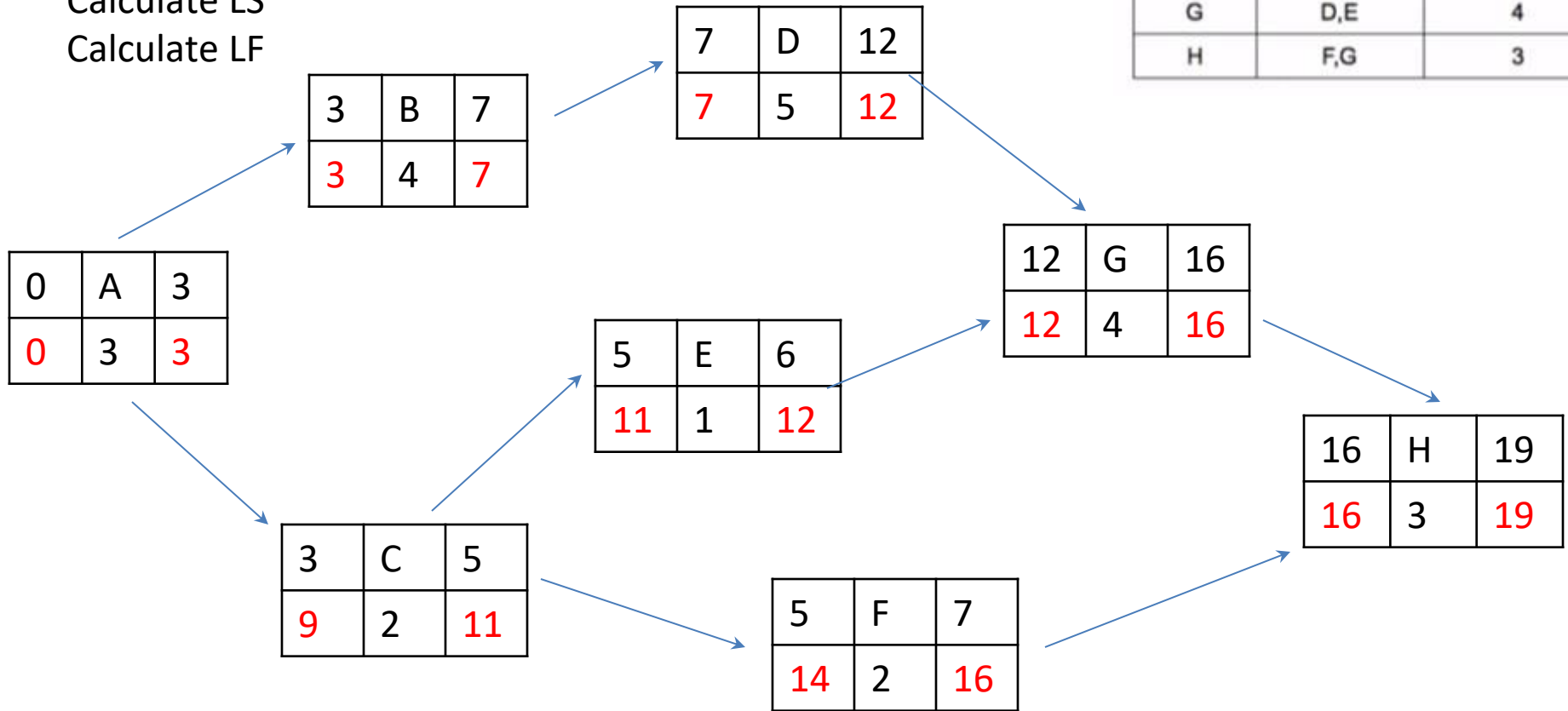
Calculate LS
Calculate LF



11 }
14 } min

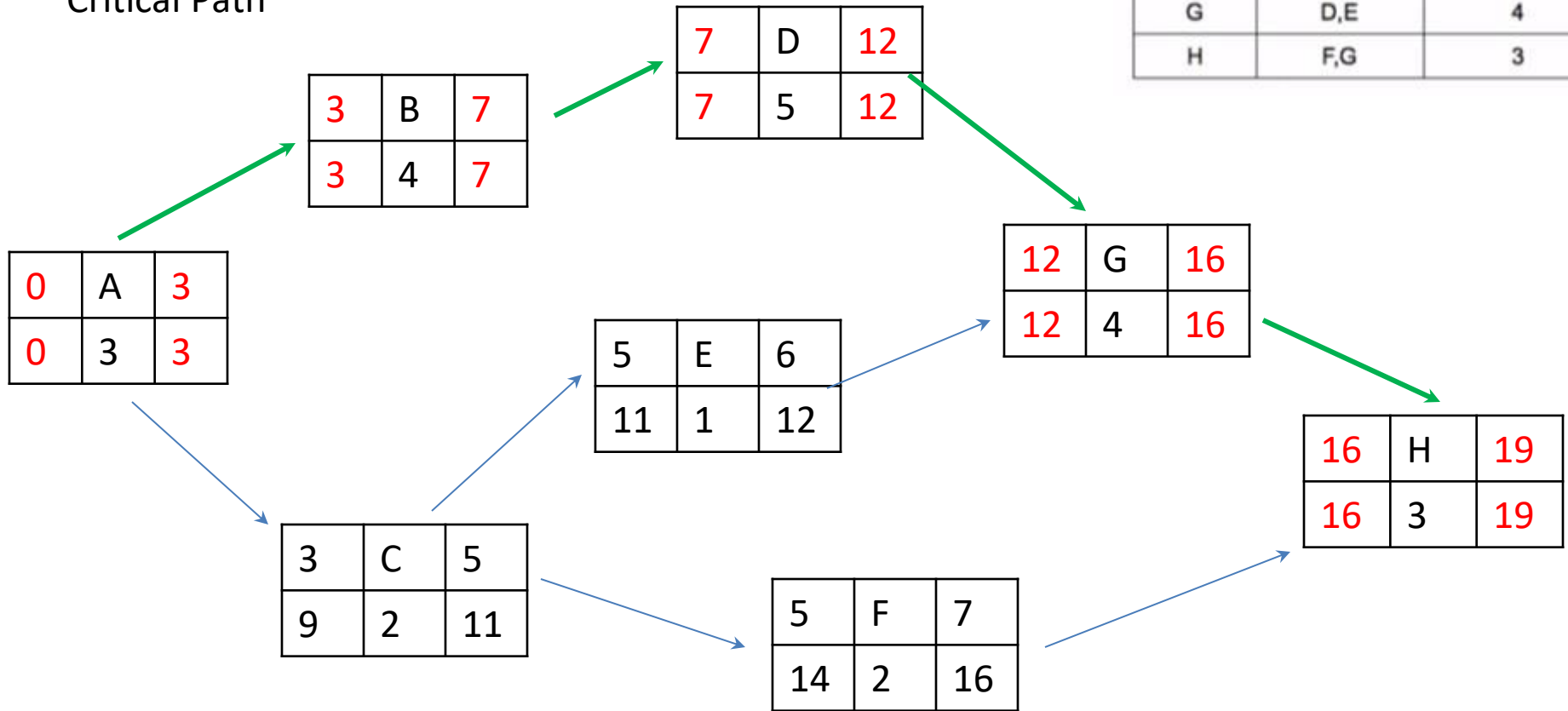
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Calculate LS
Calculate LF



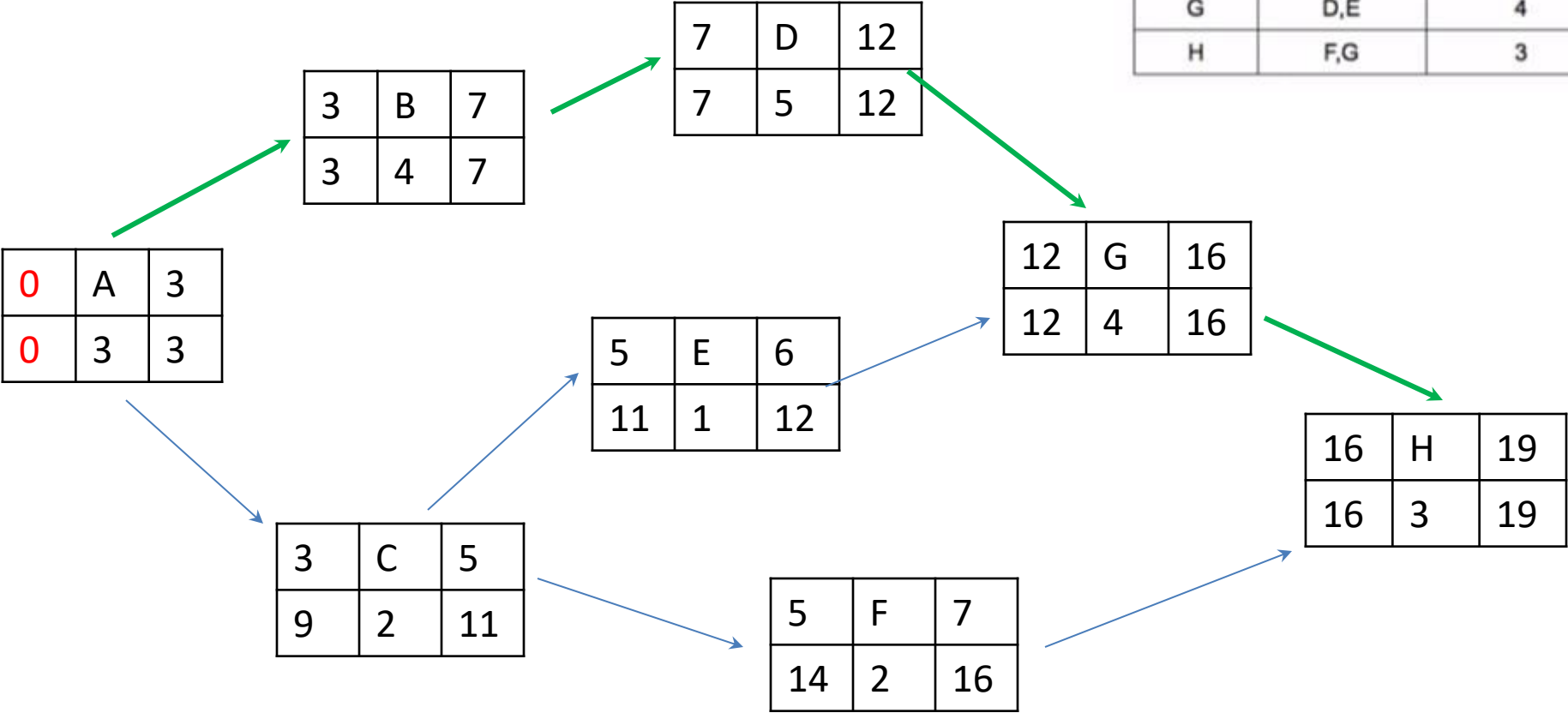
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Critical Path



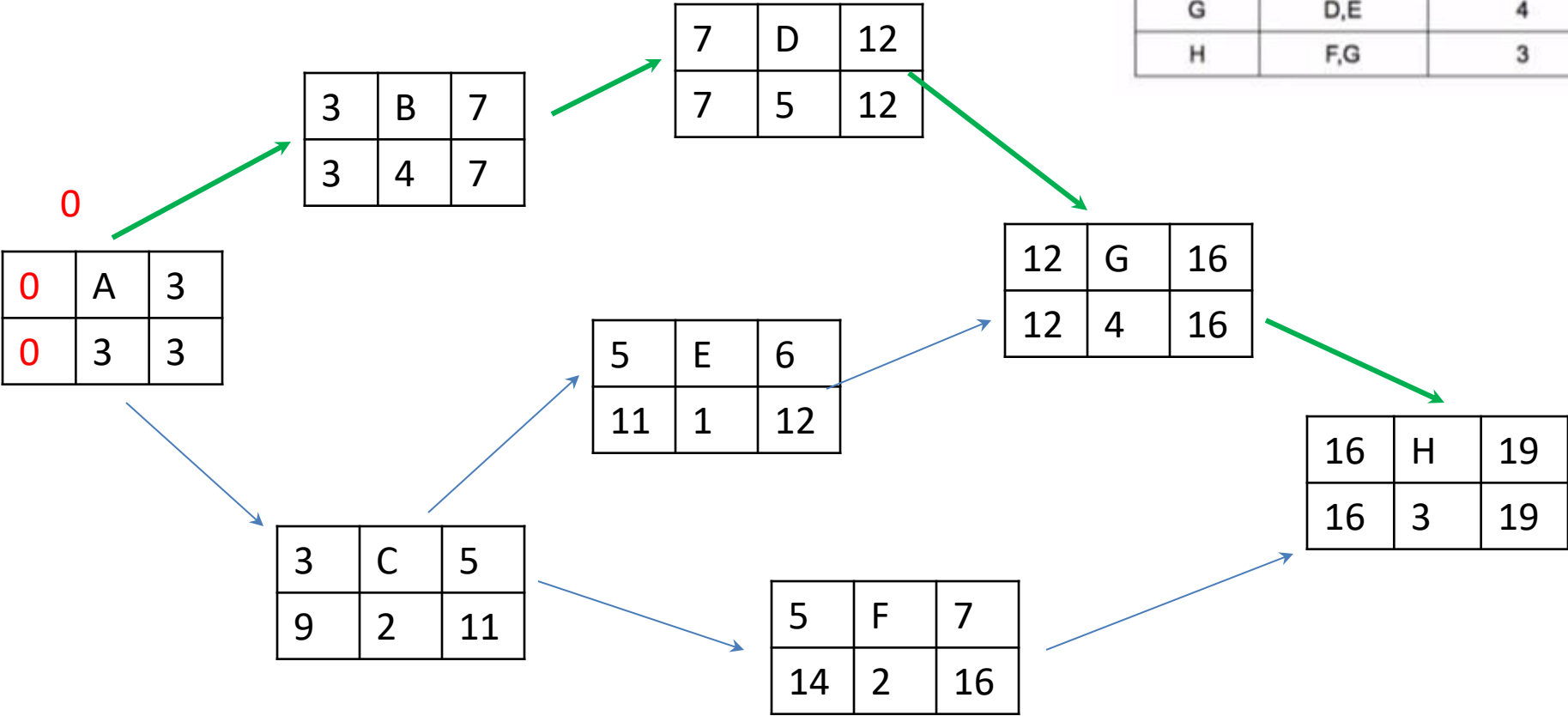
Total Float (TF)/Slack
TF=LF-EF (Finish Float)
TF=LS-ES (Start Float)

Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
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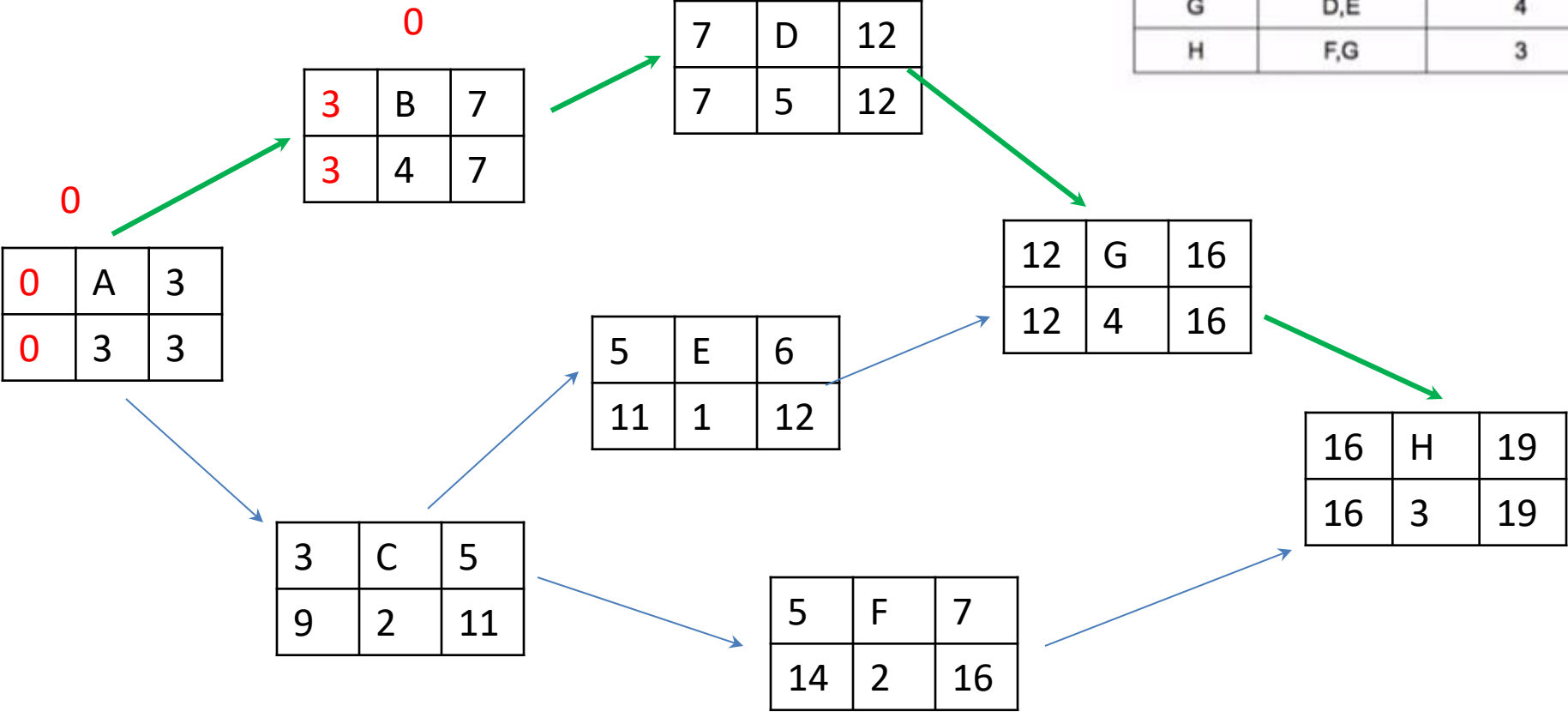
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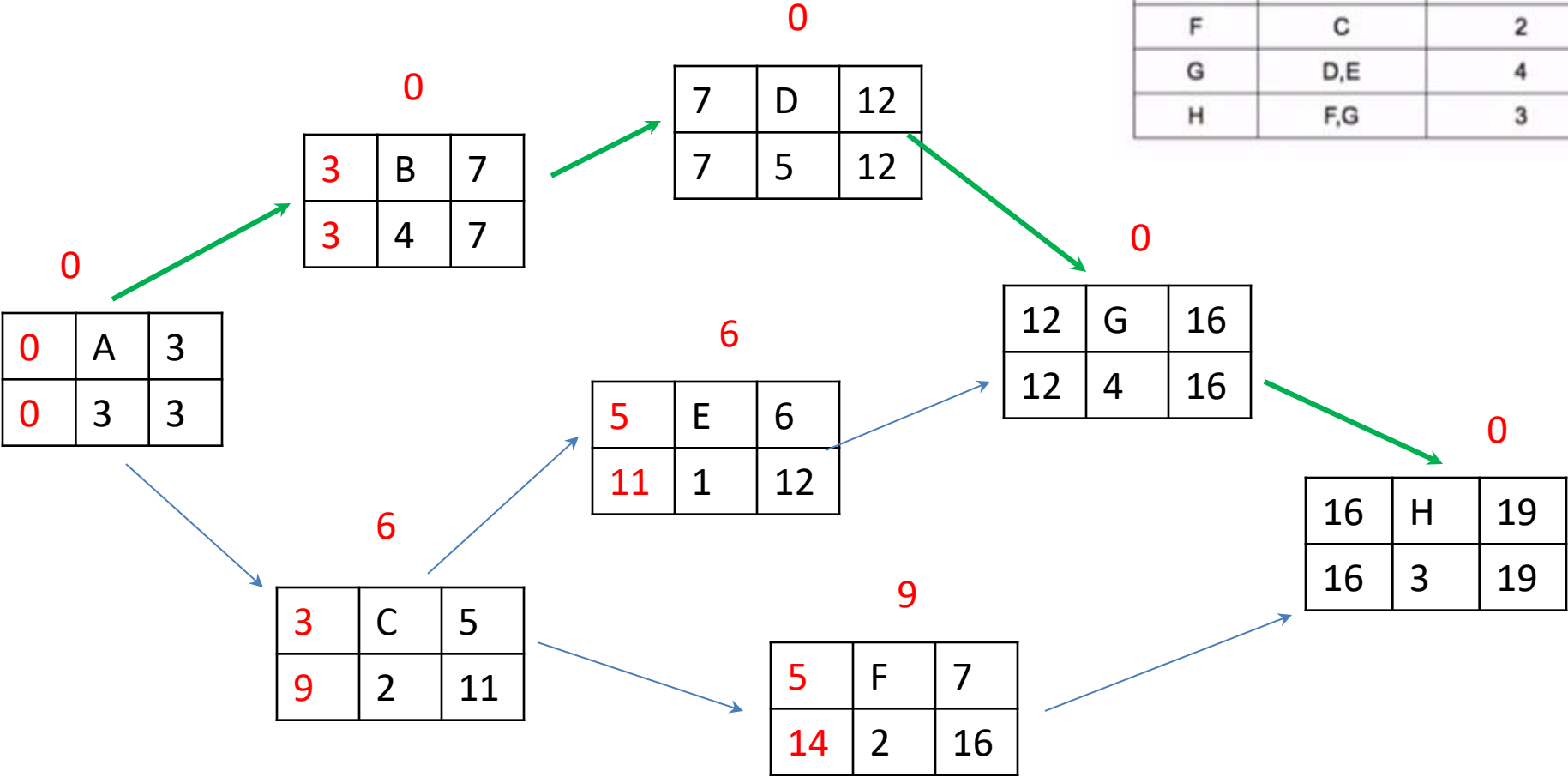
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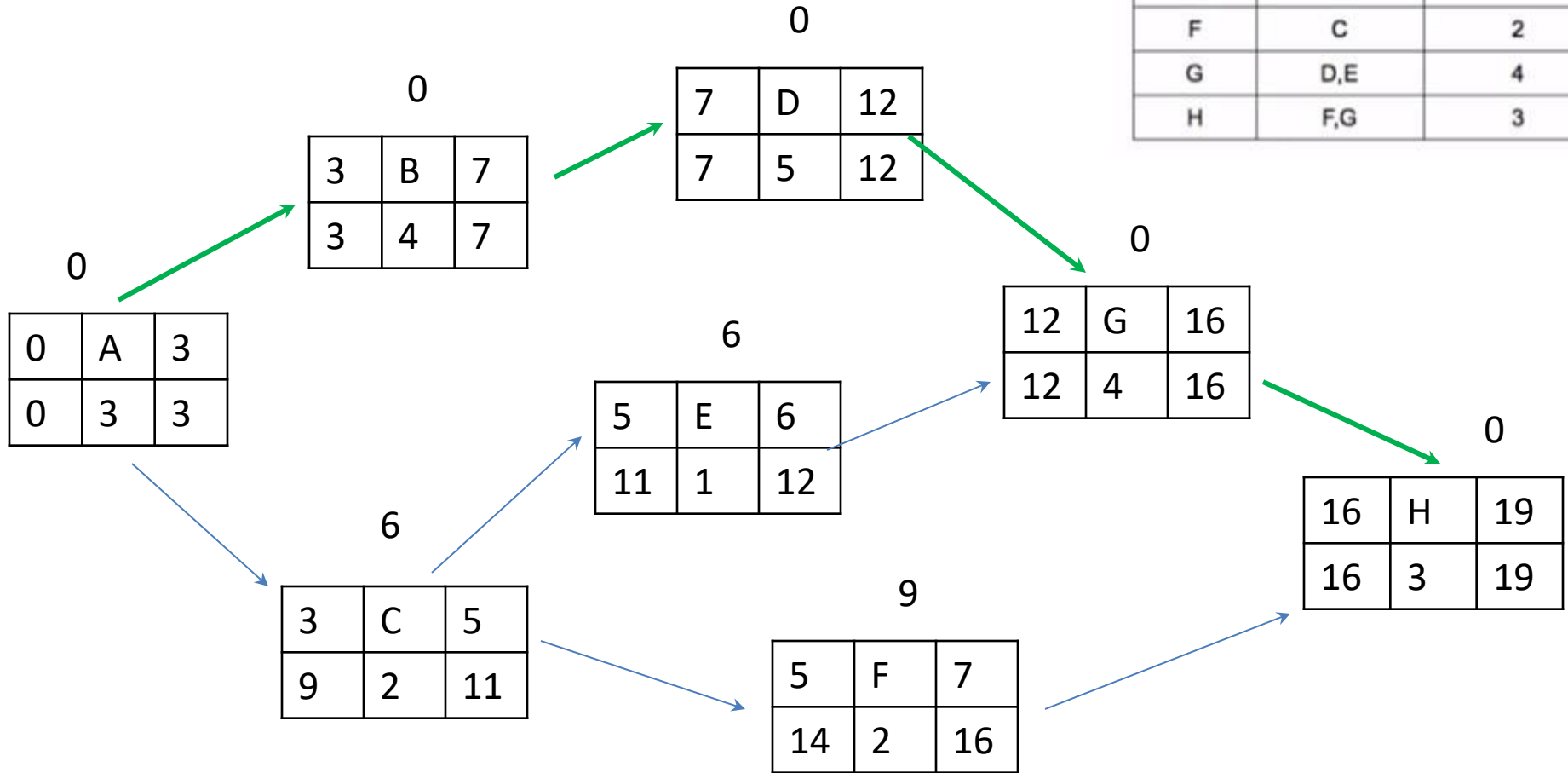
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G	D,E	4
H	F,G	3



Free Float (FF)

$FF = \text{Min}(\text{of ES of Successors}) - \text{ES}(\text{Activity}) - \text{Duration}(\text{Activity})$

Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
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G	D,E	4
H	F,G	3



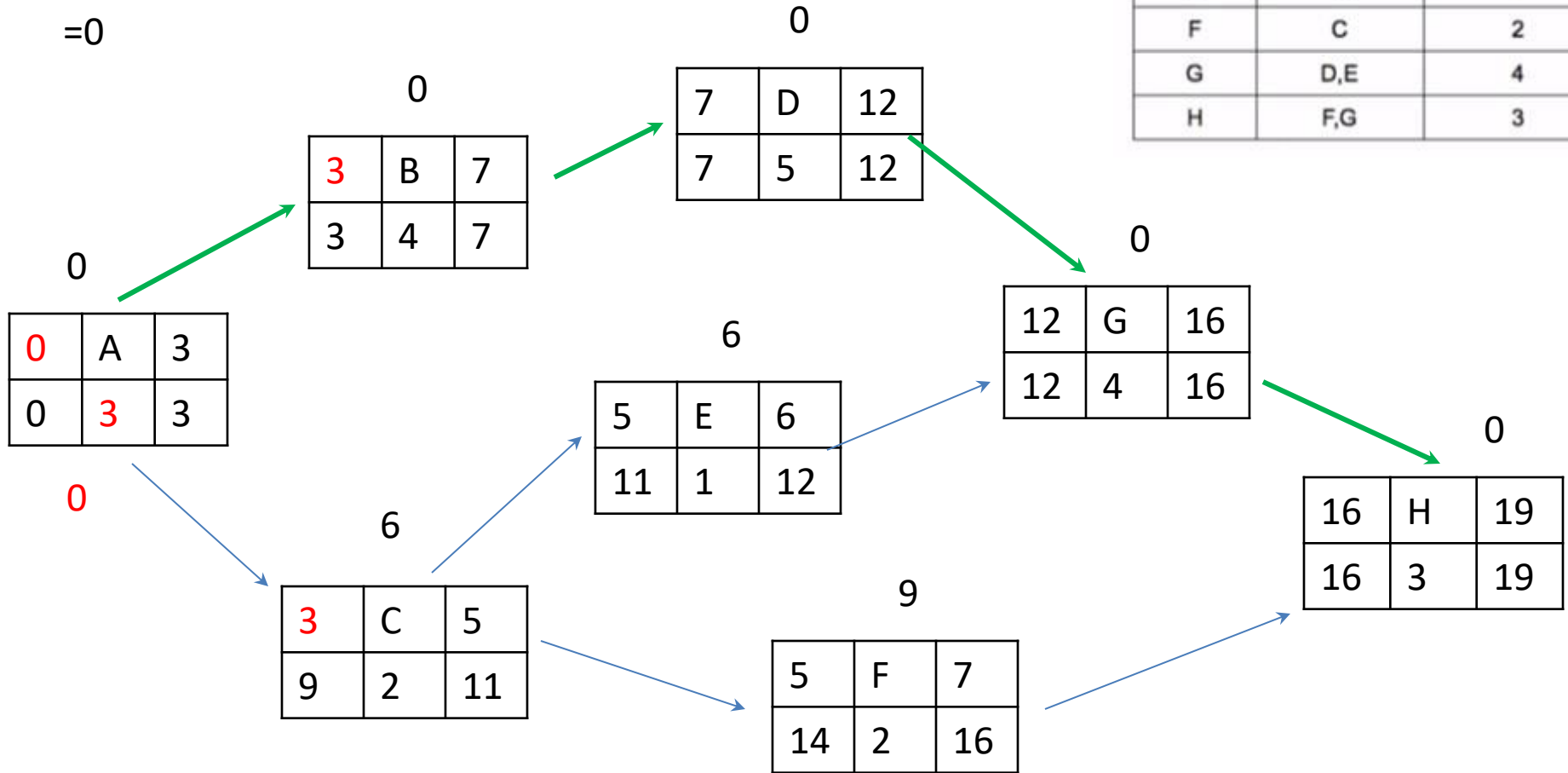
Free Float (FF)

FF=Min(of ES of Successors)-ES(Activity)-Duration(Activity)

$$=3-0-3$$

$$=0$$

Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
F	C	2
G	D,E	4
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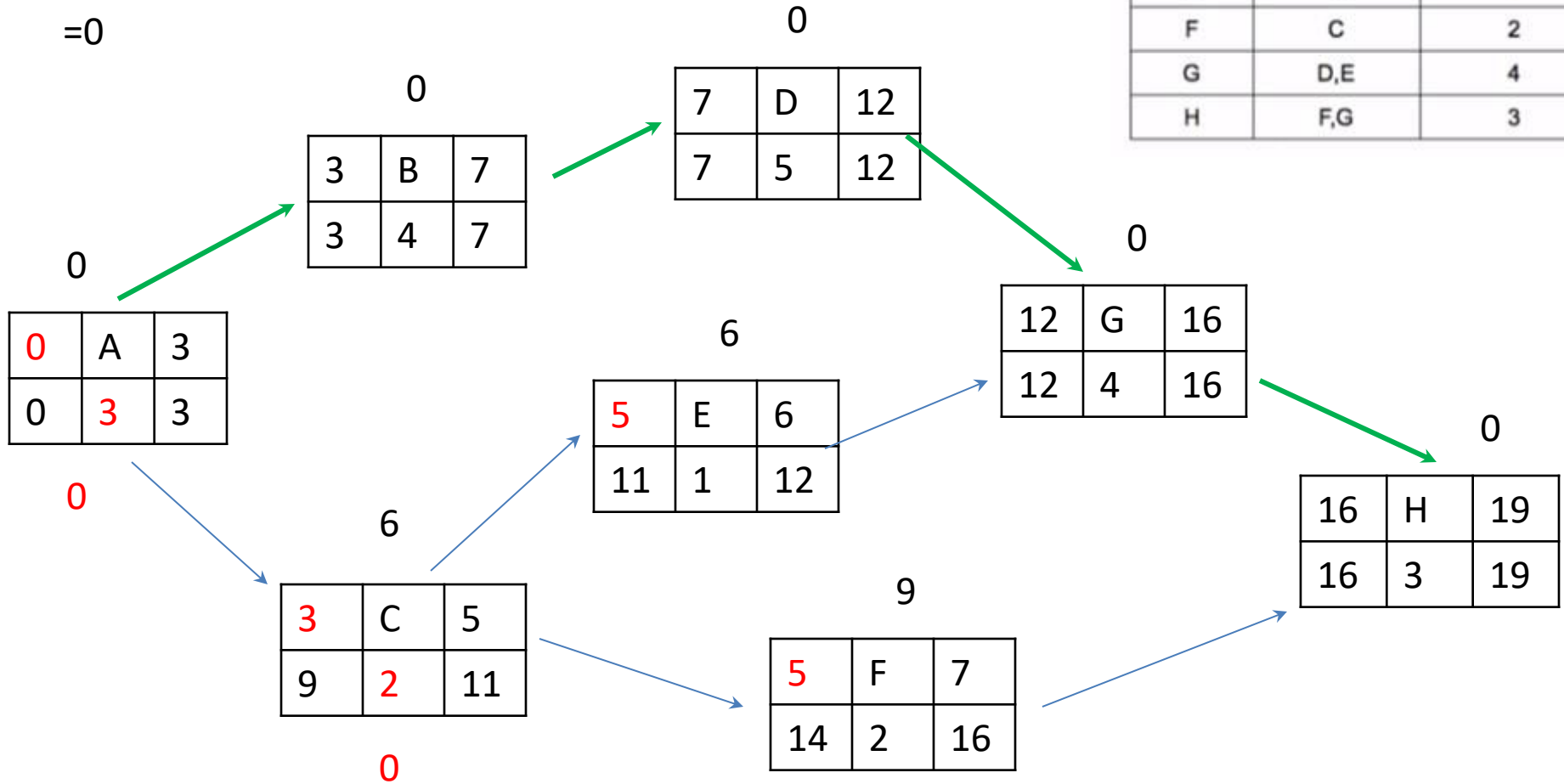
Free Float (FF)

$FF = \text{Min}(\text{of ES of Successors}) - \text{ES}(\text{Activity}) - \text{Duration}(\text{Activity})$

$$= 5 - 3 - 2$$

$$= 0$$

Activity	Predecessor	Duration (days)
A	-	3
B	A	4
C	A	2
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E	C	1
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G	D,E	4
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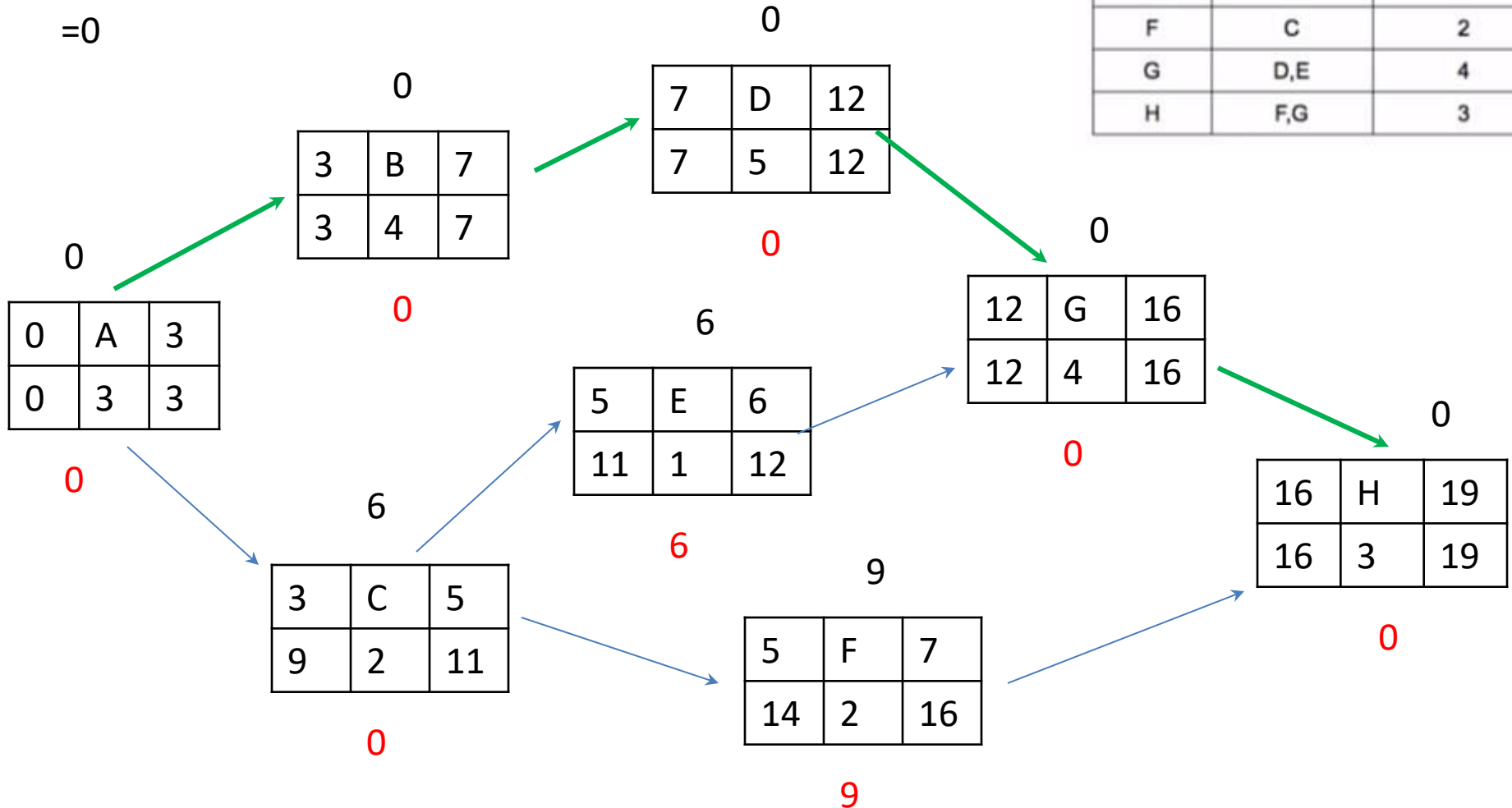
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$$= 5 - 3 - 2$$

$$= 0$$

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GANTT CHART

- ❑ A GANTT chart is a type of **bar chart** that illustrates a **project schedule**.
- ❑ After the PERT/CPM analysis is completed, the following phase is to construct the GANTT chart and then to re-allocate resources and re-schedule if necessary.
- ❑ GANTT charts have become a common technique for representing the phases and activities of a project **work breakdown structure**.
- ❑ It was introduced by Henry Gantt around 1910 – 1915.

GANTT CHART

❓ Characteristic

- ❑ The bar in each row identifies the corresponding task
 - ❑ The horizontal position of the bar identifies start and end times of the task
 - ❑ Bar length represents the duration of the task
 - ❑ Task durations can be compared easily
 - ❑ Good for allocating resources and re-scheduling
 - ❑ Precedence relationships can be represented using arrows
 - ❑ Critical activities are usually highlighted
 - ❑ Slack times are represented using bars with dotted lines
 - ❑ The bar of each activity **begins** at the activity **earliest start time (ES)**
 - ❑ The bar of each activity **ends** at the activity **latest finish time (LF)**.
-

GANTT CHART

☐ Advantages

- ☐ Simple
- ☐ Good visual communication to others
- ☐ Task durations can be compared easily
- ☐ Good for scheduling resources

☐ Disadvantages

- ☐ Dependencies are more difficult to visualise
- ☐ Minor changes in data can cause major changes in the chart

CONSTRUCTING GANTT CHART

- The steps to construct a GANTT chart from the information obtained by PERT CHART and CPM are:
 1. Schedule the critical tasks in the correct position.
 2. Place the time windows in which the non-critical tasks can be scheduled.
 3. Schedule the non-critical tasks according to their earliest starting times.
 4. Indicate precedence relationships between tasks.

CONSTRUCTING GANTT CHART

- ❓ Example of an early GANTT chart construction:

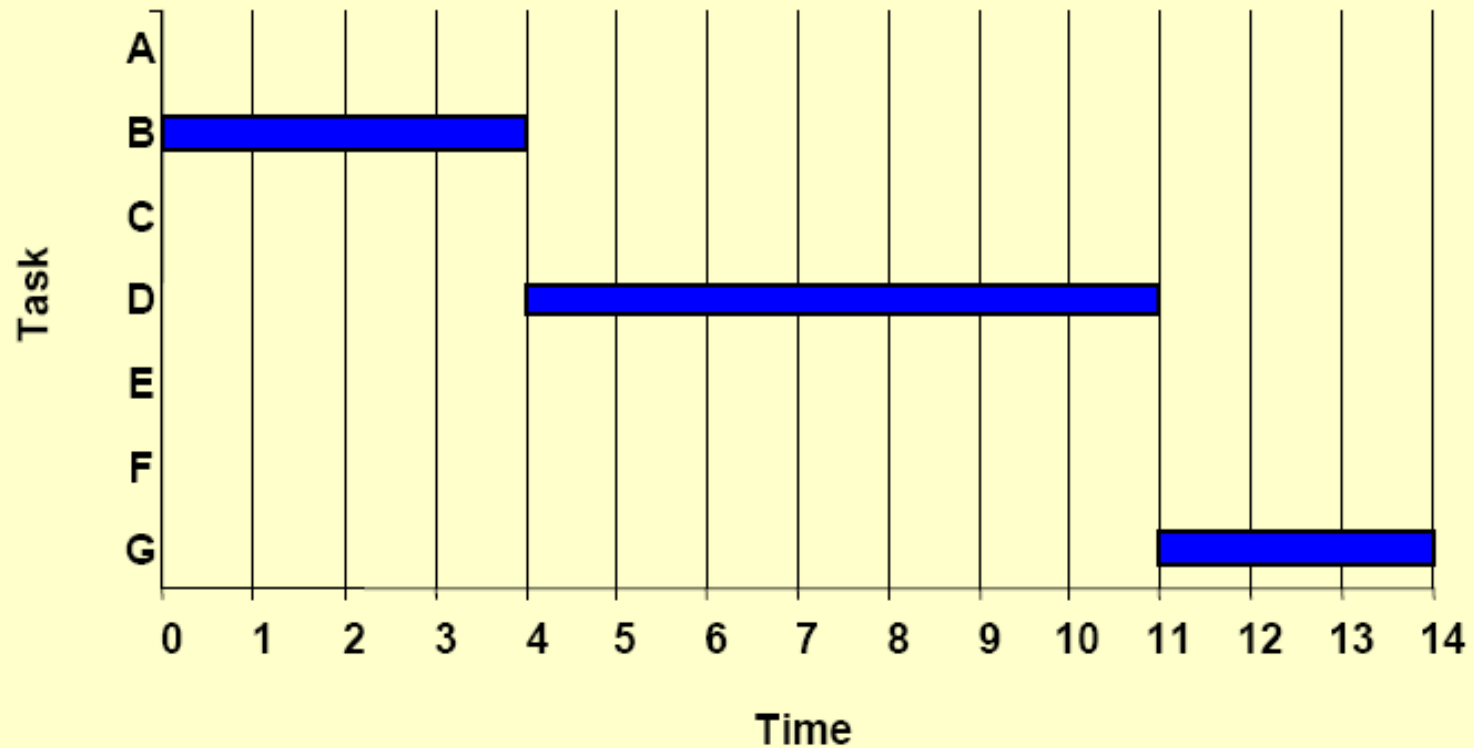
Task	Duration	Precedence	ES	EF	LS	LF	Slack Time	Critical Task
A	3		0	3	3	6	3	N
B	4		0	4	0	4	0	Y
C	5	A	3	8	6	11	3	N
D	7	B	4	11	4	11	0	Y
E	2	B	4	6	8	10	4	N
F	4	E	6	10	10	14	4	N
G	3	C,D	11	14	11	14	0	Y

CONSTRUCTING GANTT

CHART

Step 1. Schedule critical

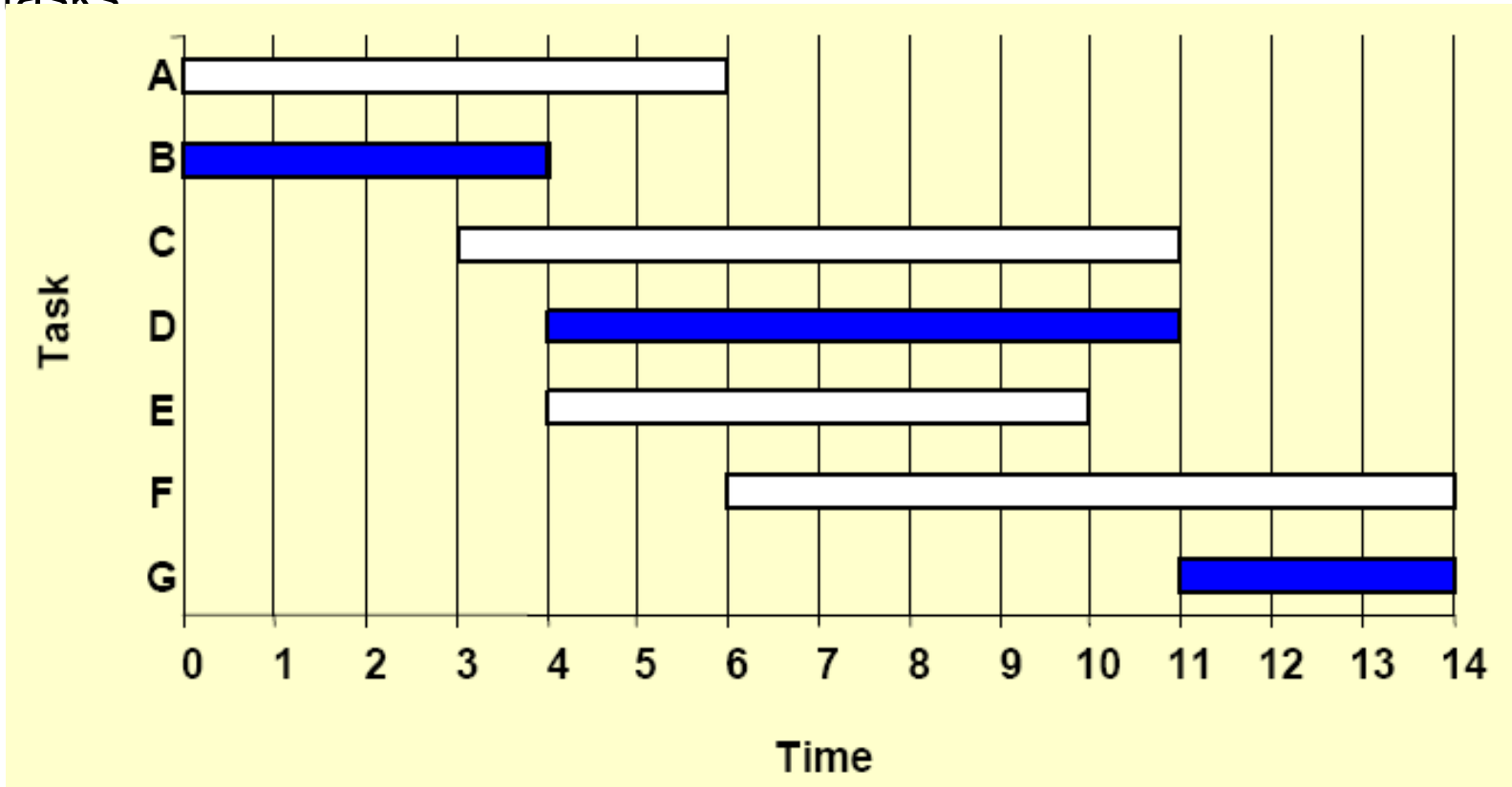
tasks



CONSTRUCTING GANTT

CHART

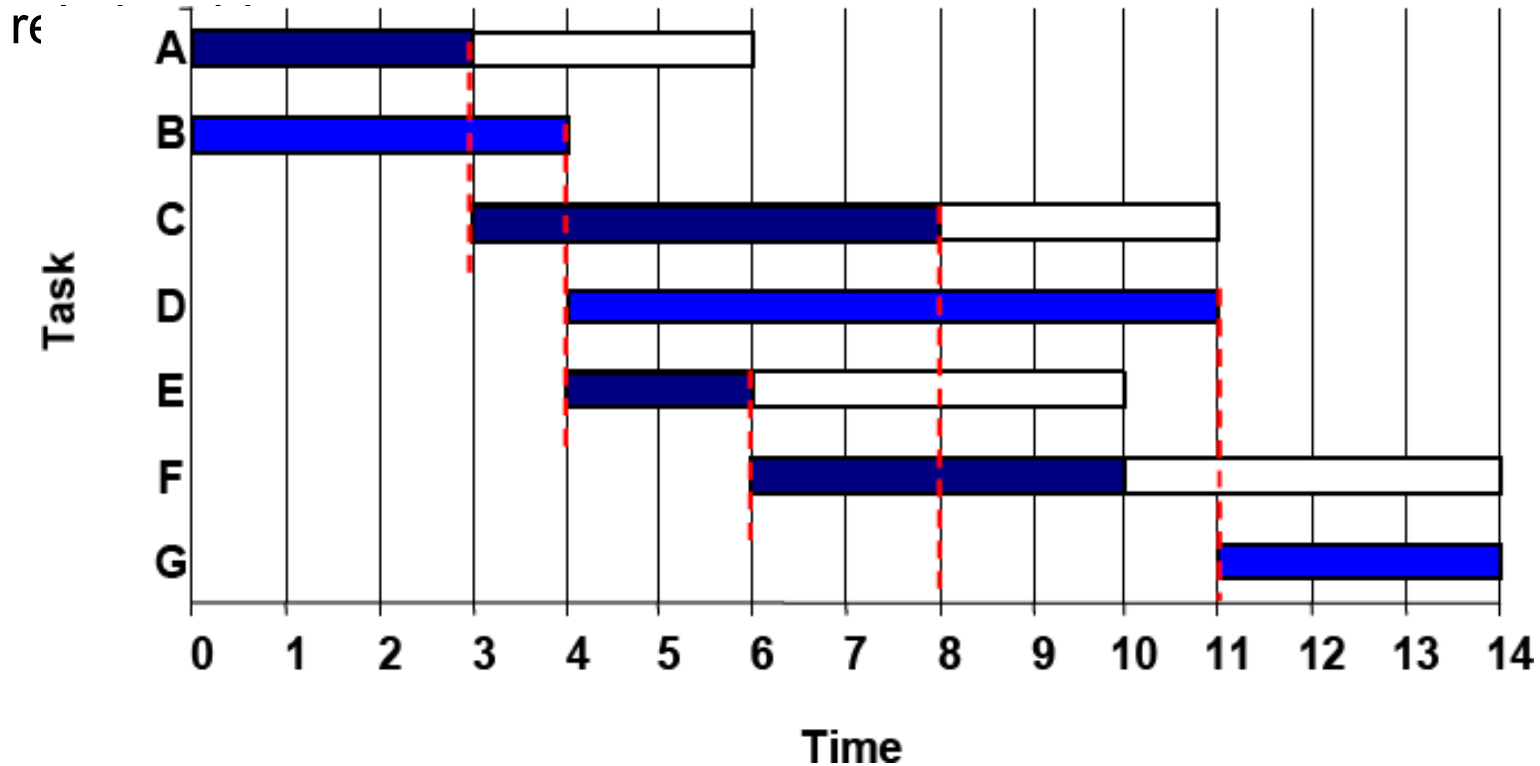
Step 2. Place time windows for non-critical tasks.



CONSTRUCTING GANTT CHART

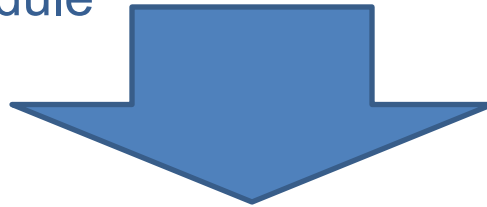
Step 3. Schedule non-critical tasks

Step 4. Indicate precedence



STAFFING & RE-SCHEDULING

- Once the project schedule, (*e.g. GANTT chart*), has been constructed, take into account
 - available staff hours
 - slack times and
 - the project schedule



Assign staff and other resources to each activity in the project

STAFFING & RE-SCHEDULING

- ❑ **Resource Smoothing** is a technique used to *re-allocate resources and re-schedule activities*.
- ❑ In resource smoothing, **non-critical tasks** are **re-scheduled** within their time window.
- ❑ **Staff Utilization:** $(\text{duration of activity} \times \text{staff required for each activity, all added together}) / (\text{maximum staff required} \times \text{duration of project})$

STAFFING & RE-SCHEDULING

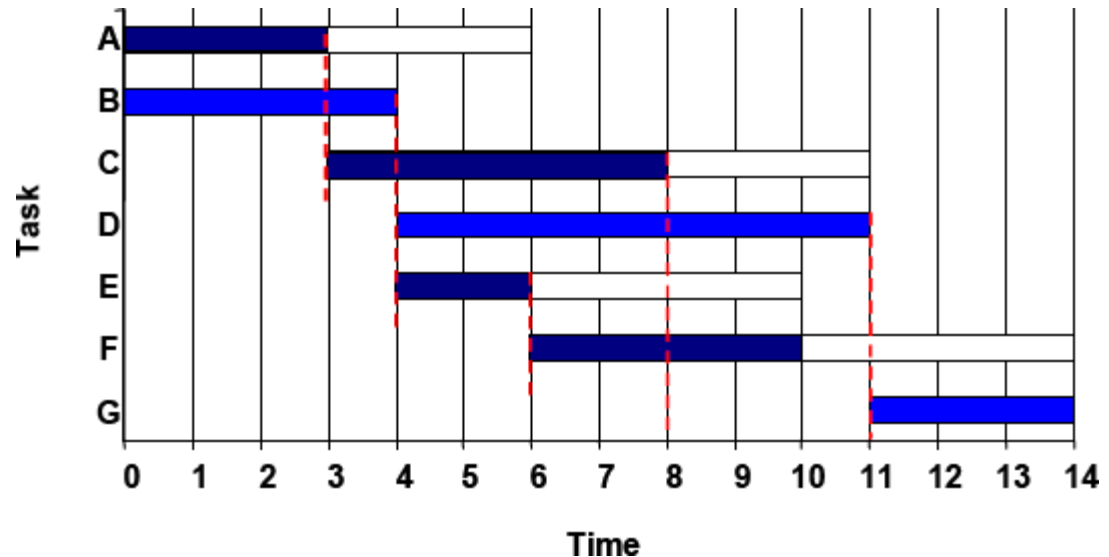
Example 1

Task	Duration	Precedence	ES	EF	LS	LF	Slack Time	Critical Task
A	3		0	3	3	6	3	N
B	4		0	4	0	4	0	Y
C	5	A	3	8	6	11	3	N
D	7	B	4	11	4	11	0	Y
E	2	B	4	6	8	10	4	N
F	4	E	6	10	10	14	4	N
G	3	C,D	11	14	11	14	0	Y

STAFFING & RE-SCHEDULING

The original schedule (constructed above) for this project is as shown below.

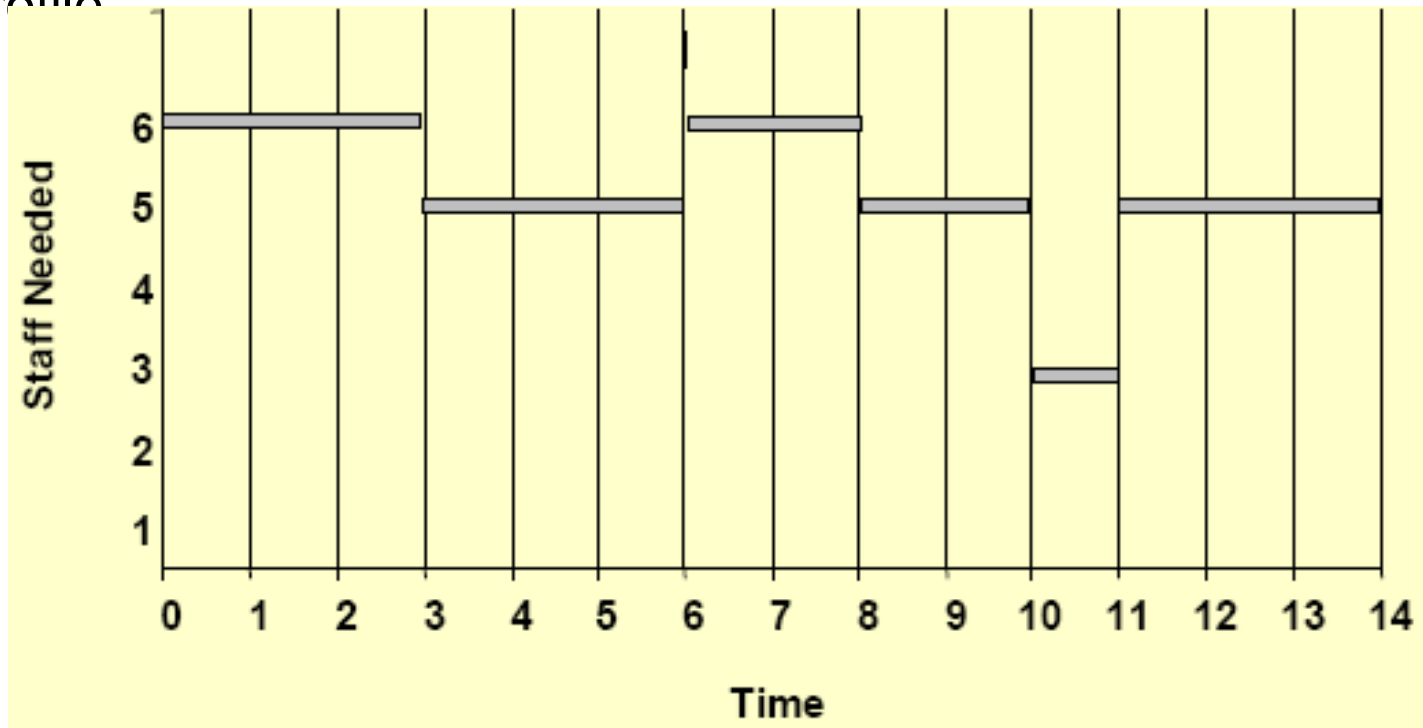
Task	Duration	Staff Needed
A	3	2
B	4	4
C	5	1
D	7	3
E	2	1
F	4	2
G	3	5



1. Staff utilisation = $(3 \times 2 + 4 \times 4 + 5 \times 1 + 7 \times 3 + 2 \times 1 + 4 \times 2 + 3 \times 5) / (14 \times 6) = 0.857$
 = 85.5%

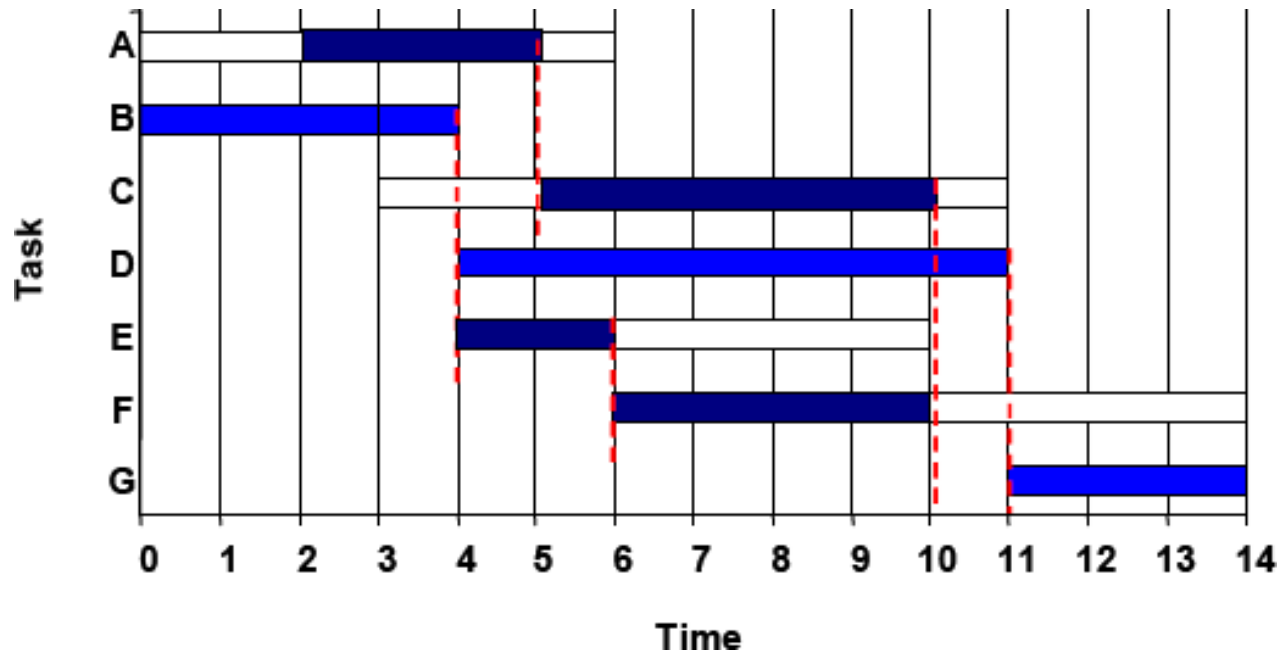
STAFFING & RE-SCHEDULING

2. Work out the Staff Profile



STAFFING & RE-SCHEDULING

- Now, assume that there are 6 people available for working in this project but one of them returns from holidays at time=2.
- So *re-scheduling* is needed because activities A and B cannot be carried out in parallel until time=2.



STAFFING & RE-SCHEDULING

- ❓ Suppose another scenario in which equipment and materials needed to carry out activities E and F are available at time=5 and time=9 respectively instead of being available at the activities ES time. Then, rescheduling is needed but the overall duration of the project is not affected.

